Artificial intelligence—enabled precision medicine for inflammatory skin diseases

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Recent advances in artificial intelligence (AI) and multimodal data collection are revolutionizing dermatology. Generative AI and machine-learning approaches offer opportunities to enhance the diagnosis and treatment of autoimmune and inflammatory skin diseases, including atopic dermatitis, psoriasis, hidradenitis suppurativa, vitiligo, alopecia areata, and rheumatic skin disease. This review examines the current landscape of AI applications for inflammatory skin diseases and explores how generative AI and machine-learning methods can advance the field through deep phenotyping, characterization of disease heterogeneity, drug discovery, precision medicine, and delivery of clinical care. We discuss the promises and challenges of these technologies and present a vision for their integration into clinical practice.

Keywords: Autoimmunity, Bioinformatics, Inflammation, Inflammatory skin diseases

INTRODUCTION

The increasing availability of comprehensive patient information, spanning multimodal imaging, multiomic profiles, and real-world clinical data is driving a significant transformation in health care. Artificial intelligence (AI)—based computational methods have emerged as a powerful tool to analyze these complex datasets and extract clinically meaningful insights, enabling advances in disease diagnosis, treatment planning, and clinical decision support (Khera et al, 2023; Saab et al, 2025; Tu et al, 2025). These technological advances are particularly timely for inflammatory skin diseases, as our understanding of the immune system and inflammatory pathways continues to evolve and enable targeted therapies (Hawkes et al, 2017; van der Schaft et al, 2019). Given the complex pathophysiology and heterogeneity of inflammatory skin diseases, AI offers vast opportunities for applications ranging from understanding precise disease mechanisms to aiding in patient communication.

As a brief primer, AI refers to the general paradigm of computational systems to mimic human intelligence, whereas machine learning (ML), a subset of AI, refers to approaches for computer programs to learn from data for specific tasks. Generative AI encompasses approaches that can create or simulate information. Various learning paradigms exist, with the most common being supervised learning, in which a model learns from data that have been previously labeled with a known outcome (eg, diagnosing psoriasis, classifying atopic dermatitis [AD] severity). In contrast, the unsupervised learning paradigm enables models to discover structures or patterns in the data, enabling identification of novel disease subtypes based on molecular or clinical profiles without preconceived classifications (eg, identifying rosacea morphological subtypes). Intermediate approaches that integrate features of both supervised and unsupervised learning include semi-supervised (using small amounts of labeled data and large amounts of unlabeled data) and self-supervised (learning from unlabeled data with learned labels) learning. Finally, the reinforcement learning paradigm enables models to learn from trial and error (eg, optimizing alopecia areata [AA] patient education on the basis of feedback).

These paradigms are executed by various model architectures. Dermatology, with its strong reliance on visual cues and images, has flourished along with advances in computer vision and image analysis, employing architectures for image-based tasks. For years, convolutional neural networks (CNNs) have been a standard choice for skin diseases, particularly with application of skin cancer diagnosis. A prominent demonstration of this was in 2017, when deep CNNs achieved performance comparable with that of dermatologists in classifying images of skin lesions (Esteva et al, 2017). A more recent architecture, the transformer, enables incorporation of both local detail and global context in data and can be utilized as generative AI to create synthetic content mimicking the data on which they are trained. Transformer models are based on the idea of "self-attention," in which they learn context and meaning by tracking

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Abbreviations: AA, alopecia areata; ACD, allergic contact dermatitis; ACR, American College of Rheumatology; AD, atopic dermatitis; AI, artificial intelligence; AIME, Articulate Medical Intelligence Explorer; AUC, area under the curve; BP, bullous pemphigoid; CAM, class activation mapping; CLE, cutaneous lupus erythematosus; CNN, convolutional neural network; DM, dermatomyositis; EHR, electronic health record; FDA, Food and Drug Administration; HS, hidradenitis suppurativa; ICD, irritant contact dermatitis; LASSO, least absolute shrinkage and selection operator; LLM, large language model; ML, machine learning; PsAID, Psoriatic Arthritis Impact of Disease; PV, pemphigus vulgaris; SALT, Severity of Alopecia Tool; SLE, systemic lupus erythematosus; SSc, systemic sclerosis; STAT, signal transducer and activator of transcription; ViT, vision transformer

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relationships in sequential data such as words in a sentence. These transformers power current state-of-the-art models for images and text, including large language models (LLMs) and vision transformers (ViTs). A foundation model refers to a model trained on a large dataset that can then be adapted for a specific task. Some recent relevant examples include Articulate Medical Intelligence Explorer (AIME) for differential diagnosis (McDuff et al, 2025), MONET for skin image annotation (Kim et al, 2024), and PanDerm for various applications from lesion segmentation to melanoma prognostication (Yan et al, 2025). Recently, there is a rise of agentic AI—which incorporates multiple autonomous systems capable of executing tasks independently—that enables greater task complexities (Zou and Topol, 2025). Although many existing tasks involve applications in skin cancer and melanoma, there is currently an evolving landscape for applications in inflammatory dermatology and beyond, with potential for personalized pathway targeting and management. Further multimodal data and applications include dermatopathology classification, teledermatology triaging, and beyond (Cazzato and Rongioletti, 2024; Muhaba et al, 2022; Soenksen et al, 2021).

In this paper, we comprehensively review current applications of ML and AI in inflammatory skin disorders (Figure 1). We first review broad applications in inflammatory skin diseases, including disease classification and prediction, deep phenotyping, biomarker discovery, therapeutic design, and clinical care (Figure 1). Then, we discuss disease-specific applications, including but not limited to AD, psoriasis, acne, rosacea, hidradenitis suppurativa (HS), vitiligo, AA, cutaneous lupus, dermatomyositis (DM), and morphea/scleroderma (Table 1 and Supplementary Table S1). Finally, we explore present challenges and future opportunities for future diagnostic and therapeutic development and their potential for clinical implementation.

METHODS

To identify relevant articles for this review, we conducted a comprehensive literature search of PubMed and Google Scholar, which included peer-reviewed manuscripts, preprints, and conference papers. The search was supplemented by reviewing references of identified articles and following "similar article" suggestions. The search utilized keywords including a combination of AI paradigms or models (eg, ML, AI, generative AI, foundation model, deep learning), applications (eg, prediction, classification, subtyping, clustering, phenotyping, biomarker discovery, drug repurposing), and inflammatory skin diseases (eg, AD, HS, vitiligo, AA, cutaneous lupus, etc). Final inclusion in this manuscript was based on a manual evaluation by the authors of each individual publication's quality and relevance to the topic.

AI APPLICATIONS IN INFLAMMATORY DERMATOSES

Classifying lesions and diagnosing inflammatory skin conditions

Although early AI applications and advancements in dermatology have primarily focused on melanoma detection (Choy et al, 2023; Esteva et al, 2017; Young et al, 2021), recent developments have expanded to encompass a greater diversity of skin diseases, including lesions of inflammatory skin disorders such as acne, psoriasis, eczema, rosacea, vitiligo, and other chronic inflammatory conditions. Multiple publications have focused on single-disease or multidisease classification on the basis of clinical images, dermatoscope images, and teledermatological images (Huang et al, 2021a; Li Pomi et al, 2024; Liu et al, 2020; Muñoz-López et al, 2021; Prabhu et al, 2018[1]; Wang et al, 2021b; Zhu et al, 2021b) (Table 1 and Supplementary Table S1 [first column]). Classification for both low- and high-level processing tasks may be utilized in the approach to inflammatory skin lesion identification (Choy et al, 2023; Li et al, 2023b). Low-level application includes processing of raw images or segmentation of skin lesions to highlight areas of interest for further analytics. As an example, Czajkowska et al (2022, 2021) demonstrated an automated framework for segmenting the epidermis from high-frequency ultrasound images in inflamed skin). High-level classification includes determining a diagnosis or directly predicting management, with an example from Abhishek et al (2021) demonstrating better performance with direct management prediction than inferring management through the prediction of diagnosis first. Beyond traditional images, the scope of diagnostic classification has expanded to accommodate a greater variety of input data sources, such as histological slides (Hosny et al, 2020; Muhaba et al, 2022) to nonimage data such as gene expression profile sequencing from tape strips, Raman spectroscopy, and electronic health records (EHRs) (Greenfield et al, 2023; Gustafson et al, 2017; He et al, 2021; Jiang et al, 2022) (Figure 1). Generative AI may be utilized for image enhancement and data augmentation to address dataset limitations such as low sample size and artifacts such as hair and ink markings.

Advances in diagnosis include the fine tuning of existing models to enable efficient learning on smaller datasets. By leveraging baseline knowledge from models pretrained on large datasets (such as ImageNet), fine-tuned models do not have to learn basic visual features from scratch (such as edges, textures, or shapes). Often, lower-level tasks such as edge detection and segmentation are incorporated within deep learning architectures. Hosny et al (2020) demonstrated the specialization of AlexNet for classifying 7 predefined skin lesion types such as melanoma, dermatofibroma, and vascular lesions. This approach to AI specialization is valuable for future approaches in inflammatory dermatoses given less need for extensive new datasets or computational resources.

A particularly useful development is the emergence of interpretable AI that allows for clinicians to scrutinize how models are making decisions. Approaches include those that probe into attention networks and gradient-based saliency maps, such as class activation mapping (CAM), to highlight and enhance our understanding of specific features such as morphology and pigmentation. Mohan et al (2025) showed improved performance of self-supervised transformer approaches such as DINOV2 in classifying a wide array of skin diseases and utilized techniques such as Grad-CAM and SHAP for model interpretation. Corbin and

[1] Prabhu V, Kannan A, Ravuri M, Chablani M, Sontag D, Amatriain X. Prototypical clustering networks for dermatological disease diagnosis. arXiv 2018.

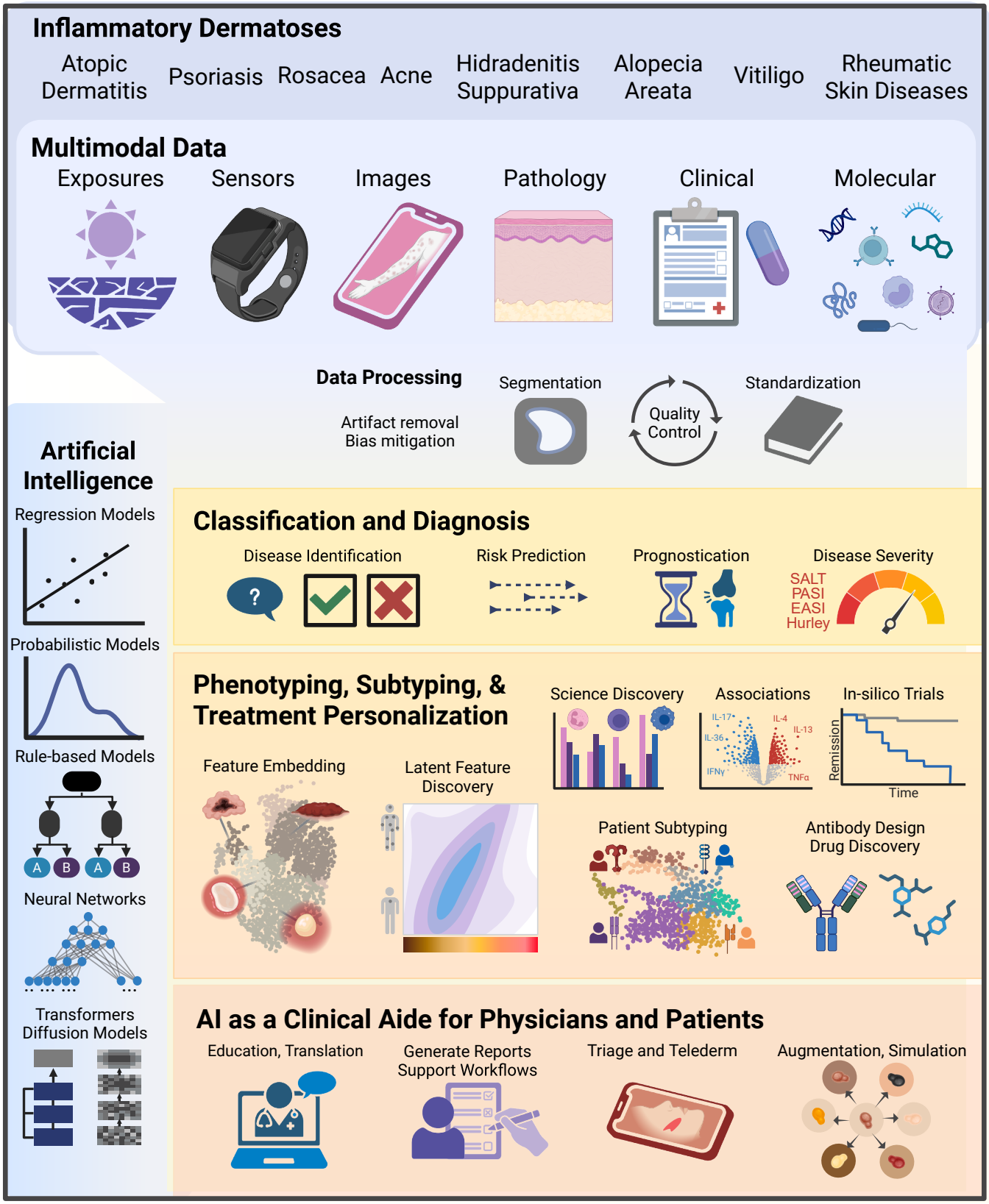

Figure 1. Landscape of current and future applications of AI to advance precision medicine for inflammatory skin diseases. This figure summarizes the landscape of AI in inflammatory skin disease, starting from inflammatory dermatoses at the top. Data from external and internal factors encompassing various modalities are shown in the subsequent row. The raw data may undergo preprocessing, such as segmentation and quality control, to prepare for subsequent tasks. The left column reveals the range of AI approaches and algorithms for various applications in the figure discussed in this review. Classification applications utilize known labels or outcomes for a specific task, such as diagnosis, prognosis, or prediction of disease severity. Exploratory tasks may include using unsupervised learning to identify features (or an attribute) that are directly or indirectly quantified (eg, latent features). This may then lead to iterative

Table 1. AI-Enabled Advancements across the Spectrum of Inflammatory Skin Diseases

Diseases	Application		
	Classification and Diagnosis	Phenotyping, Subtyping, and Treatment Personalization	AI for Workflows and Patient Care
Atopic dermatitis	13	16	14
Psoriasis	14	14	13
Rosacea	4	10	3
Acne	4	3	3
Hidradenitis suppurativa	3	3	0
Alopecia areata	3	4	0
Vitiligo	2	2	0
Cutaneous lupus erythematosus	3	5	0
Morphea or scleroderma	1	4	1
Dermatomyositis	1	3	0
Other or multiple	23	23	20

Abbreviation: AI, artificial intelligence.

Presented are numbers of reviewed studies grouped by skin disorder in rows and grouped by application in columns. Studies are outlined in Supplementary Table S1.

Marques (2023) utilized Grad-CAM to assess skin-type bias and showed improved fairness metrics with joint regularization and transformer-based synthetic data blending. This integration of transparency is crucial for building clinical trust and can also offer insights into disease morphology from the model's perspective. These classification and prediction approaches are particularly useful for primary care referral decisions or prognostication for clinical decision support for both general practitioners and dermatologists.

Phenotyping, subtyping, and treatment personalization

In addition to improving diagnosis, ML methods can serve to expand our understanding of inflammatory skin diseases, enabling applications such as disease characterization, biological hypothesis generation, and precision treatment approaches (Félix Garza et al, 2019; Tang et al, 2024). Biomarker identification and mechanistic insights can be elucidated with evolving computational approaches that integrate diverse data types, such as genomics, transcriptomics, cell surface markers, and clinical phenotypes, to enable the understanding of patterns and relationships across modalities (Clayton et al, 2021; Jiang et al, 2022; Wang et al, 2024c). This can be valuable in complex conditions such as systemic lupus erythematosus (SLE), scleroderma, morphea, and other autoimmune disorders, where target inflammatory pathways and subtypes have been identified that may reflect underlying patient or molecular heterogeneity and can help inform targeted treatment or prognosis (Andreoletti et al, 2021; Choi et al, 2023; Cutts et al, 2024). Clustering algorithms at either the data level or patient level can enable the characterization of disease subtypes based on variations in molecular mechanisms, treatment response, and prognosis (Table 1 and Supplementary Table S1 [second column] and Figure 1) (Berna et al, 2020; Cazzaniga et al, 2021; Neely et al, 2022). On a broader level, integration with population-level information can elucidate risks from external exposures (eg, pollution, humidity) as well additive risks due to comorbidities (eg, diabetes mellitus, mental health disorders). For example, there have been studies that identify external or comorbid factors influencing irritant dermatitis and HS (Fortino et al, 2020; Hua et al, 2021; Papa et al, 2023). Currently, these methods enable researchers, epidemiologists, and clinicians to try improving the understanding of disease mechanisms.

Efforts on treatment are also underway to study the integration of protein–drug interaction networks that can further accelerate therapeutic target identification and drug repurposing efforts. For instance, Patrick et al (2018a) used a word-embedding-based ML approach to identify candidate drugs for AD and psoriasis on the basis of enriched genes from lesional skin transcriptomic data. Mathematical models of biological interactions have also been employed to increase the understanding of biological processes and drug response (Barraza et al, 2024; Kardynska et al, 2023). These approaches analyze drug response pathways to predict treatment efficacy for specific disease subtypes, whereas transcriptomic analyses can provide deeper insight into disease mechanisms as another approach to identify targets for drug development (Yamanaka et al, 2023) (Table 1 and Supplementary Table S1 [second column]). Moreover, AI approaches are also investigated at the stage of clinical trial design and treatment selection through patient stratification based on disease severity, subtypes, or predicted responses (Huang et al, 2023; Hurault et al, 2020). The potential for generative AI has also been proposed for applications such as biomarker identification and drug

improvement in a supervised ML task or aid in discovery tasks (eg, cell surface marker associations, cytokine associations) and precision medicine approaches (eg, patient subtyping based on receptors). In silico means through computer modeling or simulation. The bottom rows show how generative AI can integrate with research and clinical care to improve patient phenotyping and development of new treatments for inflammatory skin disease. AI, artificial intelligence; EASI, Eczema Area and Severity Index; ML, machine learning; SALT, Severity of Alopecia Tool.

design (Shanehsazzadeh et al, 2023[2]; Ying et al, 2024[3]). The generative models enabling these advancements vary widely, encompassing approaches from foundational physical equations and probability distributions to sophisticated neural network architectures such as transformers (including LLMs), generative adversarial networks, and diffusion models (Figure 1 [left column]). The integration of various learning paradigms and diverse data types enables the discovery of complex relationships among biological mechanisms, environmental exposures, and clinical manifestations. Although exploratory, the potential to identify targeted therapeutic candidates (eg, IL-4 inhibitors, Jak inhibitors) can support the advancement of precision medicine strategies that target disease-relevant pathways in inflammatory skin diseases. These applications are still at the beginning stages, with gaps and opportunities for further expansion.

AI as a clinical aide for physicians and patients

AI is beginning to reshape clinical workflows for clinicians and care delivery for patients, with new uses in dermatology emerging. Across medical specialties, the development of AI scribes and clinical summarizers can enhance clinical efficiency by automating documentation and supporting workflows (Miao et al, 2025; Williams et al, 2024). The accelerated adoption of telehealth after the COVID-19 pandemic also enhanced teledermatology advancements, spurring investigations into how generative AI can aid with virtual consultations, enhance remote image assessment, and facilitate accurate documentation (Shapiro and Lyakhovitsky, 2024; Vodrahalli et al, 2023, 2021) (Figure 1). This technological advancement holds significant promise for expanding access to dermatological expertise, especially for patients in underserved or remote areas (Muñoz-López et al, 2021).

AI is also helping clinicians with image interpretation and clinical reasoning. ViT models are also being developed to generate explanations or interpretations of dermatological images directly (Lin et al, 2025; Mohan et al, 2025; Zhou et al, 2024a). For example, SkinGEN from Lin et al (2025) proposes a framework for combining vision language models with generative AI to visually explain dermatological diagnoses. Kim et al (2024) introduced an image–text foundation model, MONET (medical concept retriever), trained on dermatology image–text pairs from literature with capability of annotating both clinical and dermatoscope images. Yan et al (2025) introduced PanDerm, a multimodal vision foundation model with various abilities, including providing a differential diagnosis that incorporated inflammatory skin diseases and predicting disease progression. Recently, AI has been shown to be capable of clinical reasoning and able to provide differential diagnosis that is noninferior to that of expert dermatologists and with incorporation of visual data, as seen from Google's AIME that can analyze smartphone skin images (Saab et al, 2025[4]). Although AI applications to aid clinicians in dermatology care are promising, applications in inflammatory skin diseases remain an area with significant opportunity (Table 1 and Supplementary Table S1 [right column]).

In terms of applications in patient education, generative AI can improve communication by aiding in translation or creating accessible explanations of medical information, such as pertaining to chronic management of inflammatory dermatoses or side effects from steroids (Aydin et al, 2024; Johri et al, 2025). AI-generated translations of medical notes and reports aim to make complex information more understandable for patients. For example, Zhang et al (2024b) assessed ChatGPT-4's ability to translate hypothetical dermatopathology reports into patient-friendly language. Although physician raters found the translations generally favorable, they also noted occasional errors for longer, more complex reports (Zhang et al, 2024b). One study also evaluated the utility of LLMs in answering questions from patients, which can be useful to guide patients because there is an increasing prevalence of LLM use for health information among lay people (Mendel et al, 2025). These studies are still in the beginning stages, and dedicated research and development will be needed to harness its potential for complex challenges provided by inflammatory skin diseases.

DISEASE-SPECIFIC APPLICATIONS OF AI

AD

Classification and diagnosis. Recent approaches extend image-based CNN models (Hammad et al, 2023; Wu et al, 2020), with newer applications emerging that synthesize multimodal data to advance the understanding of AD. For instance, a supervised ML classifier for AD based on gut transcriptomics and microbiome identified 35 predictive genes (including *GRP1*, *CCL22*, *TTC27*) and 50 microbiota features (such as *Akkermanisia spp*, *Verrucomicrobia spp*, *Propionibacterium spp*) that help distinguish patients with AD from healthy controls (Jiang et al, 2022). AD classification models with microarrays identify genes related to epidermal development, keratinocyte function, immune responses, and pyroptosis (Ghosh et al, 2015; Wu et al, 2023). Besides distinguishing AD from healthy skin, classification models have been applied to differentiating AD from psoriasis. Liu et al (2022b) used single-cell RNA sequencing and cellular indexing of transcriptomes and epitopes by sequencing on CD45+ immune cells, highlighting genes linked to T helper 2/T helper 17 balance, especially in T-resident memory cells. He et al (2021) similarly utilized RNA sequencing from tape strips to distinguish between AD, psoriasis, and controls, identifying improved classification with dendritic and T-cell markers (CD3, ITGAX/CD11c, CD83), downregulated genes involved with terminal differentiation (*FLG2*, *LCE5A*), genes with T helper 2 skewed in AD tissue (*IL13*, *CCL17/TARC*, *CCL18*) or T helper 17/T helper 1–related expression in psoriasis (IL-17A, IL-36A, IFN-γ, CXCL9), and expression of

[2] Shanehsazzadeh A, McPartlon M, Kasun G, Steiger AK, Sutton JM, Yassine E, et al. Unlocking *de novo* antibody design with generative artificial intelligence. bioRxiv 2023.

[3] Ying W, Wang D, Hu X, Qiu J, Park J, Fu Y. Revolutionizing biomarker discovery: leveraging generative AI for bio-knowledge-embedded continuous space exploration. arXiv 2024.

[4] Saab K, Freyberg J, Park C, Strother T, Cheng Y, Weng WH, et al. Advancing conversational diagnostic AI with multimodal reasoning. arXiv 2025.

nitric oxide synthase 2 and inducible nitric oxide synthase (He et al, 2021). Overall, the models confirm a skewed T helper 2 profile in AD versus a T helper 17/T helper 1 profile in psoriasis. These studies show how ML models can illuminate novel biological pathways involved in the pathogenesis of AD and how they are similar or distinct from other inflammatory skin diseases.

Models trained on EHR data, which include least absolute shrinkage and selection operator (LASSO) logistic regression and natural language processing or transformer-based patient representation with XGBoost model, have successfully classified AD (Gustafson et al, 2017; Wang et al, 2024b). Other promising modalities for classification include handheld confocal Raman spectroscopy, with one study classifying patients with AD with spectroscopic signatures that depict stratum corneum proteins and lipids (Dev et al, 2022) and another that can assess AD severity on the basis of measured water, ceramide, and urocanic acid content (Ho et al, 2020). With regard to environment-related risk, Huang et al (2021b) conducted a 14-year follow-up birth cohort study involving 1439 mother−infant pairs with models trained on combined demographic, environmental (eg, air pollutant), and allergen data and noted prenatal exposure to nitrogen dioxide as a significant predictor of prenatal AD.

Phenotyping, subtyping, and treatment personalization. Approaches to identify biomarkers and subtypes of AD are performed with the goal of analyzing personalized treatment approaches and responses. AD severity models include a Bayesian inference model using prior AD flares and treatment response (Hurault et al, 2020) and a gradient boosting and regression model from clinical data, identifying severity-associated factors such as later onset, total IgE levels, male sex, and atopic stigmata (Maintz et al, 2021). Wearable biosensors can assess AD severity and treatment response by integrating various technologies: actinography to quantify itch, sensors for physical properties such as skin capacitance and temperature, and surface imaging and chemical sensors that capture inflammatory markers such as changes in vascularity and pH (Greenfield et al, 2023; Khan et al, 2024; Yang et al, 2025).

Besides AD severity phenotypes, unsupervised clustering has enabled identification of subphenotypes that associate with disease clearance and enable personalization of treatment and prediction of disease outcomes. Berna et al (2020) clustered baseline clinical data from over 8000 children in the Pediatric Eczema Elective Registry and identified 5 distinct AD subphenotypes, defined by factors such as race, age of onset, and allergy profiles, with statistically significant differences in long-term disease clearance. Another study trained models on EHR and claims data to predict dupilumab nonresponse, identifying factors such as concomitant ibuprofen use and higher comorbidity scores as predictors for nonresponse at 6 months, offering potential flags for closer monitoring or alternative management approaches (Wu et al, 2022a).

Further efforts at treatment personalization involve targeted drug prediction and development. For instance, Clayton et al (2021) clustered 406 skin transcriptome samples from patients with AD, identifying 3 unique keratinocyte activation programs and therapy-dependent modifications. For instance, they identified that cyclosporine primarily affected keratinocytes responding to IL-17A, whereas dupilumab specifically reversed the IL-4− and IL-13−responsive keratinocyte profile (Clayton et al, 2021). With regard to targeted drug development, Wang et al (2022) combined network analysis, deep learning, and molecular simulation, confirming TNFα and IL-4 as key disease targets and further identifying caffeoyl malic acid as a dual target inhibitor for the treatment of AD. A couple of other efforts in AD include computational approaches for drug repositioning, particularly targeting pathways such as Jak−signal transducer and activator of transcription (STAT), immune factors, or inflammatory cytokines beyond IL-4/IL-13 (Prakash et al, 2020; Prasannanjaneyulu et al, 2022). These studies have led to the clinical development of upcoming therapies, including biologics targeting IL-31 and OX-40/OX-40L, many of which are undergoing clinical trials.

AI as a clinical aide for physicians and patients. As the daily management of chronic inflammatory skin diseases extends beyond the clinic, LLMs are increasingly playing a role in medical care. Two studies assessed LLM responses to common questions around AD, overall finding answers to be mostly acceptable but otherwise incomplete or deviated from evidence-based medicine (Lakdawala et al, 2023;Sulejmani et al, 2024).

Summary. AI applications in AD are moving beyond image-based classification to incorporate multimodal data for more nuanced diagnosis and analysis. Initial applications leverage supervised learning paradigms with image-based CNN models for lesion identification, diagnosis, and severity determination. Subsequent models aim to diagnose AD with data from transcriptome, microbiome, clinical data, or spectroscopy. A few models that are specifically trained to distinguish AD from psoriasis with gene expression highlight T helper 2 skew in AD. Unsupervised learning paradigms aim to stratify patients on the basis of severity or treatment response. In clinical care, generative models are beginning to find use as patient-facing educational aides.

Psoriasis vulgaris

Classification and diagnosis. Multiple image-based applications have been applied to psoriasis that range from segmentation to diagnosis. As an example low-level task, Shrivastava et al (2017) developed a Bayesian model for segmenting psoriatic lesions from clinical images by classifying each pixel, followed by secondary extraction of color, entropy, and light intensity to determine psoriasis severity. On a higher level, multiple studies extend deep learning CNN architectures to distinguish psoriasis from controls (Aijaz et al, 2022; Chakraborty et al, 2024; Zhao et al, 2020), determine PASI (Huang et al, 2023; Okamoto et al, 2022; Schaap et al, 2022; Shrivastava et al, 2016), or predict severity changes over time (Moon et al, 2024). Applications also include distinguishing psoriasis from potential mimics, such as AD, using images (Wu et al, 2020) and gene expression datasets (He et al, 2021; Liu et al, 2022b; Seremet et al, 2024) or distinguishing scalp psoriasis from seborrheic dermatitis using dermoscopic images (Yu et al, 2022). Beyond

diagnostics, ML applications in psoriasis care include prediction of complications such as psoriatic arthritis, plaque burden, or treatment response (Yu et al, 2020). Patrick et al (2018b) utilized genetic markers to predict the risk of developing psoriatic arthritis versus cutaneous-only psoriasis. Another prospective study on 158 patients with psoriatic arthritis demonstrated that random forest models can predict minimal disease activity at subsequent visits with strongest predictors of global pain, Psoriatic Arthritis Impact of Disease (PsAID) score, patient global assessment of disease, and Health Assessment Questionnaire Disability Index (Queiro et al, 2022).

Phenotyping, subtyping, treatment personalization. AI with molecular datasets is driving advancements in psoriasis phenotyping and treatment response. For biomarker discovery, Zhou et al (2024b) utilized LASSO and SVM-RFE ML models to identify 4 potential diagnostic biomarkers of psoriasis (UGGT1, CCNE1, MMP9, and ARHGEF28), with experimental validation supporting CCNE1 as a key candidate owing to its elevated expression in psoriatic epidermis and relevant immune cells (CD4+ T cells, neutrophils). In terms of molecular subtyping, multiple approaches utilize unsupervised learning paradigms to identify potential subtypes in psoriasis that may differentially respond to treatment. Zhang et al (2024a) clustered skin transcriptomes identifying 4 distinct psoriasis subtypes (immune activated, stromal, and intermediate) with different predicted treatment responses: the immune and intermediate subtypes favored methotrexate or anti−IL-12/23 therapies, whereas the stromal subtype appeared more responsive to anti-TNFα/anti−IL-17RA agents. Li et al (2023a) similarly identified 2 psoriasis subtypes with distinct immune infiltration profiles and potential candidate treatment: cluster 1 with higher activated CD4+ T cells and proposed glycogen synthase kinase inhibitor treatment and cluster 2 with higher plasmacytoid dendritic cell and neutrophils with proposed mTOR and phosphoinositide 3-kinase inhibition. Another study that clustered word embeddings identified top 5 drug repurposing candidates that include budesonide, hydroxychloroquine, leflunomide, mesalamine, and cyclophosphamide (Patrick et al, 2018a). Unsupervised learning paradigms also investigate psoriasis treatment−specific responses. Damiani et al (2020) utilized unsupervised artificial neural networks on baseline clinical and blood count data to predict responders to secukinumab, whereas Pournara et al (2021) utilized finite mixture models on patients with psoriatic arthritis to identify 7 clinical clusters that show variable response trajectories to different secukinumab dosing. Tomalin et al (2020) developed models with bagging and ensemble methods on blood protein data to predict 12-week treatment response for tofacitinib and etanercept, finding predictive proteins, including IL-17A/C.

AI as a clinical aide for physicians and patients. Within clinical workflows, Klçoğlu et al (2025) assessed ChatGPT's responses to frequently asked questions about psoriatic arthritis, finding that treatment-related answers scored lower than symptom-related ones. Other applications of generative AI in psoriasis are limited.

Summary. Similar to AD, psoriasis vulgaris has been a focus of AI research, particularly with regard to lesion segmentation, disease classification, and assessment of disease severity (eg, PASI, PsAID). Most approaches utilize supervised learning paradigms with CNN deep learning models to analyze images. Recent advances focus on treatment personalization based on selection of targeted therapies (eg, anti−IL-12 vs anti−IL-17) or predicting therapeutic outcomes (eg, secukinumab response).

Rosacea

Classification and diagnosis. A few studies developed image-based deep learning models to classify rosacea, such as RosNet (adapted from Inception-ResNet-v2) and FACES (an ensemble deep learning model) (Binol et al, 2020; Park et al, 2023). Some deep learning models are trained to differentiate rosacea from potential mimics (acne, seborrheic dermatitis, eczema) and classify rosacea subtypes (erythematotelangiectatic, papulopustular, and rhinophymatous), with one study highlighting dermoscopic features such as vascular polygons and yellow and red halos to differentiate rosacea from other facial inflammatory diseases (Ge et al, 2022; Zhao et al, 2021).

Phenotyping, subtyping, and treatment personalization. Current approaches leveraging AI to understand rosacea pathogenesis includes investigating molecular relationships and shared diagnostic phenotypes. From genetics data, Gao et al (2025) utilized bidirectional 2-sample Mendelian randomization analysis, finding a potential causal link between rosacea and cardiovascular disease, whereas Deng et al (2023) identified variants in LRRC3, SH3PXD2A, and SLC26A8, with subsequent functional analysis supporting a neurogenic inflammation mechanism in rosacea. Wang et al (2024a) investigated immune mechanisms linking rheumatoid arthritis and rosacea, identifying and validating CXCL19 and CCL27 as shared biomarkers.

Unsupervised clustering methods have been utilized to identify molecular endotypes of rosacea. From transcriptomics data, Mao and Li (2024) identified 2 rosacea subtypes (neurogenic and inflammatory), and a subsequent linear regression model on 4 genes (*PNPLA3*, *CUX2*, *PLIN2*, *HMGCR*) can diagnose neurogenic rosacea with an area under the curve (AUC) of 0.90 on a validation cohort. With respect to drug design for rosacea, Barraza et al (2024) identified downregulated SKAP2 and upregulated S100A7A as potential disease-related targets, which they then confirmed through computer simulations, showing stable binding with existing rosacea drugs (eg, isotretinoin, permethrin) as well as possible repositioned drugs (eg, dasatinib, lovastatin).

AI as a clinical aide for physicians and patients. Unsupervised learning paradigms have been utilized to understand patient topic categories from online forums (Rajalingam et al, 2023) as well as clinical practice patterns for medication prescriptions (Nicholas et al, 2024). No related generative AI publications were found for rosacea.

Summary. AI applications in rosacea have primarily focused on diagnosis and distinguishing from clinical mimics, particularly through deep learning models for image

analysis. Recent work includes the use of unsupervised clustering paradigms to explore the molecular heterogeneity of rosacea for treatment personalization. This demonstrates how ML models are able to identify new clinical features to improve clinical diagnosis of rosacea, identify new drug targets, and reposition existing drugs for the treatment of rosacea.

Acne vulgaris

Classification and diagnosis. Acne lesion identification is sometimes included in multiclass diagnostic models. For example, Karthik et al (2022) utilized a modified attention-based CNN to differentiate acne from psoriasis, actinic keratosis, and melanoma.

Phenotyping, subtyping, and treatment personalization. Current approaches are mainly aimed at assessing acne severity, with approaches that include an ensemble pruning framework (Liu et al, 2022a) and transfer learning with an image-based CNN model (Yang et al, 2021).

AI as a clinical aide for physicians and patients. Lakdawala et al (2023) investigated ChatGPT's responses to patient questions around acne, identifying lower accuracy responses concerning acne treatment. Other applications of generative AI in acne are overall limited.

Summary. Compared with that in other inflammatory dermatoses, AI applications in acne vulgaris are primarily focused on severity assessment. Diagnostic and patient support applications are limited.

HS

Classification and diagnosis. Within the last few years, ML has been utilized to assist in the diagnosis of HS. From claims data, Ali et al (2025) identified HS with XGBoost model and identified top features that include the use of sulfamethoxazole/trimethoprim, use of clindamycin phosphate, and having an obesity diagnosis. Kirby et al (2024) trained boosting algorithms to diagnose HS and distinguish it from mimics such as cutaneous abscesses and cellulitis, finding the strongest predictive features of age, sex, and risk factors (obesity, skin infections, skin infection treatments). For severity determination, Wiala et al (2024) automated smartphone image classification of HS using CNNs, with AUC up to 0.89 and noting that classification bias was caused by the presence of tattoos and excessive hirsutism.

Phenotyping, subtyping, and treatment personalization. Cazzaniga et al (2021) performed cluster analysis on clinical data of 1100 patients with HS and identified distinct early-onset and late-onset clusters, with the latter group having significantly lower ORs of factors such as current smoking, neck involvement, and family history of the disease. González-Manso et al (2021) also utilized a clustering approach on select clinical, molecular, and genetic data to propose 2 distinct endotypes of HS of similar disease severity: cluster 1 represented nonobese males with nodular lesions posteriorly, higher serum IL-10, and presence of gamma-secretase allelic variants, whereas cluster 2 represented individuals with obesity with anterior lesions, more sinus tracts and abscesses, and higher serum IL-1, CRP, IL-17, and IL-6 concentrations. Nevertheless, there is still a significant opportunity to improve subphenotyping of HS heterogeneity to understand its pathogenesis as well as develop novel precision therapies for HS.

AI as a clinical aide for physicians and patients. No related generative AI publications were found for HS. Given the chronic and devastating nature of the disease, there is a large opportunity for applications in chronic management and daily treatment optimization.

Summary. Given the clinical heterogeneity and negative impact of diagnostic delays in HS, AI is emerging as a tool for earlier identification and patient stratification, with potential to improve disease understanding for treatment personalization.

AA

Classification and diagnosis. Limited studies evaluating AA severity and progression risk exist. Lee et al (2020) developed AloNet, a deep learning framework to segment images of hair loss for AA and calculate Severity of Alopecia Tool (SALT) scores from standardized photographs. Zhang and Nie (2022) analyzed gene expression to predict progression to alopecia universalis, with XGBoost performing best for progression risk and identifying 4 key genes associated with disease progression (*CD28*, *HOXC13*, *KRTAP1-3*, *GPRC5D*).

Phenotyping, subtyping, and treatment personalization. There are a few example applications of phenotyping AA and developing treatments. Xiong et al (2023) analyzed gene expression data from transcriptomic AA datasets to identify diagnostic markers for severe AA, identifying 4 immune-monitoring genes with diagnostic effectiveness (*LGR5*, *SHISA2*, *HOXC13*, and *S100A3*). Zhang and Nie (2022) evaluated key AA progression genes with the Connectivity Map database to identify azacitidine and anisomycin as possible repurposed therapies for AA. Chen et al (2021) analyzed transcriptomics from clinical trial biopsies using regulatory network analysis to define distinct drug mechanisms of response for treatments such as ruxolitinib and tofacitinib. By matching pretreatment biopsy profiles to these inferred networks, the authors proposed that these molecular profiles can guide patient–drug matching in clinical trials (Chen et al, 2021). Generative AI paradigms have also been applied to drug discovery. Specifically, the drug discovery company Absci has demonstrated the potential for de novo antibody design using generative AI. They developed an antibody (ABS-201) targeting the prolactin receptor for androgenic alopecia, which showed improved hair regrowth in mouse models compared with minoxidil, although these developments are still in early stages (Absci, 2025).

AI as a clinical aide for physicians and patients. No high-quality studies were identified.

Summary. AI applications are primarily focused on automating severity assessment (eg, SALT) or predicting progression to alopecia totalis or universalis and the identification of novel biomarkers of disease prognosis and progression. Deep learning and generative AI models also show promise in early therapeutic development for AA, but there still remains significant opportunity for research in this area.

Vitiligo

Classification and diagnosis. Existing applications focus on image-based lesion detection and disease classification. Guo et al (2022) developed a hybrid AI model using YOLOv3 for lesion detection and UNet++ for segmentation, with better performance on Fitzpatrick skin types III/IV. Zhong et al (2024) compared Resnet and Swin Transformer deep learning models on dermoscopy images, with the latter model achieving better vitiligo classification performance (AUC = 0.94). Hillmer et al (2024) developed a CNN to quantify facial vitiligo severity with good physician agreement (interclass correlation coefficient = 0.67−0.70), with ability to track evolution over time.

Phenotyping, subtyping, and treatment personalization. Limited studies exist in vitiligo. A promising study from Wang et al (2021a) investigated potential therapeutic targets using transcriptomics data and network analysis, identifying kaempferide as a candidate drug, with proteomic profiles suggesting potential multitarget mechanism involving p38 MAPK pathway inhibition (through CDK1/PBK) and cellular reduction−oxidation homeostasis modulation to promote melanogenesis.

AI as a clinical aide for physicians and patients. No high-quality studies were identified.

Summary. AI applications have primarily focused on image-based lesion segmentation and severity assessment. Early work exploring molecular data to identify potential therapeutic interventions is emerging.

Rheumatic skin diseases

CLE: classification and diagnosis. Many AI applications in lupus primarily focus on SLE diagnosis, severity classification, or risk prediction of complications such as lupus nephritis (Ceccarelli et al, 2017; Li et al, 2022; Ma et al, 2022; Yaung et al, 2023). For example, Nair et al (2025) profiled medical records with generative AI to clinically classify SLE on the basis of American College of Rheumatology (ACR) 1997 criteria with high concurrence for the "discoid rash" criteria. A few studies specifically aim to diagnose CLE with CNN-based algorithms to classify CLE subtypes (acute CLE, subacute cutaneous lupus, discoid lupus) or differentiate from clinical mimics such as AD, psoriasis, and erythema annulare centrifugum (Li et al, 2024; Wu et al, 2021). One notable study utilized transcriptomics to classify CLE, identifying associated genes related to type 1 IFN−mediated inflammation (*TLR3*, *MYD88*, *IRF7*), STAT2 and PRF1, or IFN stimulation (such as *CXCL9*, *CCL8*, *ISG15*) (Seremet et al, 2024).

Phenotyping, subtyping, and treatment personalization. Unsupervised and supervised learning paradigms have been utilized to deconstruct CLE heterogeneity or advance its understanding relative to SLE. Among studies on lesional skin, Lee et al (2025[5]) leveraged transcriptomics to phenotype the immune heterogeneity among various SLE rashes, identifying, for example, upregulation of type 1 IFN, TNF-alpha, IL6−Jak−STAT6 pathways for subacute CLE. From peripheral blood, Tao et al (2025) combined tandem mass tag proteomics with transcriptomics, identifying IFI44 and EPSTI1 as key serum markers that can differentiate subacute CLE from SLE (with AUC = 0.85−0.89), whereas Zhu et al (2021a) utilized whole blood RNA and unsupervised k-means clustering, identifying 6 patient clusters: 2 showing elevated inflammation modules characterized by IFN, neutrophil, and cell death signatures, whereas the other 3 clusters exhibited predominant T-cell signatures.

AI as a clinical aide for physicians and patients. Nair et al (2025) profiled medical records with generative AI to classify SLE on the basis of ACR 1997 criteria. Other AI applications in CLE clinical care are limited.

Summary. AI studies in CLE aim to classify disease, differentiate from SLE subtypes, and advance the understanding of disease heterogeneity. More work is needed in the area of AI applications for phenotyping, subtyping, and drug design for lupus because limited Food and Drug Administration (FDA)−approved therapies exist. Given large amounts of available multiomics and clinical data, this represents an opportunity for AI-based computational drug repurposing and biomarker discovery.

DM

Classification and diagnosis. Limited approaches exist to diagnose DM. In one cross-sectional study, Xue et al (2023) trained a random forest model on clinical and laboratory variables to predict anti-MDA5 antibody status with AUC >0.97, finding association with high rates of fever, alopecia, periungual telangiectasia, Gottron's sign, interstitial lung disease, and arthritis.

Phenotyping, subtyping, and treatment personalization. Current AI applications leverage supervised or unsupervised learning paradigms to advance biomarker or subtype discovery. Wang et al (2024c) utilized gene expression data and weighted gene coexpression network analysis to pinpoint ISG15 as a DM biomarker, validating its biomarker potential in biopsied skin tissues. Neely et al (2022) utilized unsupervised learning methods on single-cell RNA and protein expression in blood in patients with juvenile DM, finding an inflammatory and antigen-presenting subphenotype in CD16+ monocytes and expanded transitional B-cell population as well as pan-cell−type IFN gene signature that correlated with disease activity in cytotoxic cell types. Additional ML studies into DM subtypes investigate phenotypic clustering on the basis of antibody status (eg, anti-MDA5) or predicting rapidly progressive interstitial lung disease (Koopman et al, 2024).

AI as a clinical aide for physicians and patients. No high-quality studies were identified.

Summary. AI applications in DM primarily focus on antibody prediction, biomarker identification, or subtype identification but, overall, had few studies focused on DM. There is a significant opportunity for AI approaches to improve DM disease classification, subphenotyping, and drug discovery.

[5] Lee EY, Patterson S, Cutts Z, Lanata CM, Dall'Era M, Yazdany J, et al. Transcriptomic analysis reveals immune signatures associated with specific cutaneous manifestations of lupus in systemic lupus erythematosus. bioRxiv 2025.

Morphea and scleroderma

Classification and diagnosis. Supervised learning in morphea/scleroderma have been applied to clinical data. Choi et al (2023) trained a random forest model, distinguishing 181 patients with localized morphea, generalized morphea, and systemic sclerosis (SSc), with top differentiating features (through SHAP) including anticentromere autoantibodies, anti−scl-70 autoantibodies, rheumatoid factor, histologic sclerosis, and noncutaneous manifestations. Jamian et al (2019) utilized EHRs to identify SSc, finding that the best algorithms combined clinical and billing codes, with top features including positive antinuclear antibody titers and presence of Raynaud's phenomenon keyword.

Phenotyping, subtyping, and treatment personalization. Phenotyping approaches with skin biopsies and gene expression data aim to advance disease understanding. Martínez et al (2022) analyzed lesional and nonlesional skin samples from various inflammatory skin diseases (CLE, psoriasis, AD, and SSc), revealing shared inflammatory profiles (eg, IFN, IL-12 complex) that distinguish from control skin as well as distinguishing features specific for each disease (eg, lack of IL-21 complex signature and increased TGFβ fibroblast signature in SSc). Franks et al (2019) developed an elastic net ML classifier using gene expression from SSc skin biopsies to assign to 4 intrinsic molecular subsets (inflammatory, fibroproliferative, normal like, and limited), with accuracy of 85%.

AI as a clinical aide for physicians and patients. No high-quality studies were identified.

Summary. AI applications are mainly aimed at research advancements, especially with molecular phenotyping.

Other inflammatory skin diseases

Classification and diagnosis. Beyond the previously highlighted inflammatory dermatoses, classification approaches are being extended to other related conditions such as autoimmune bullous disorders and vasculitides. For autoimmune bullous disorders, Capurro et al (2024) compared models for automatic classification of direct immunofluorescence patterns, finding that the Swin Transformer generative model performed the best across the pattern classes (intercellular, linear, negative). Among vasculitides, an AI model achieved high diagnostic performance for predicting Kawasaki disease and intravenous Ig resistance from health data (Omar et al, 2025), whereas a hierarchical attention network model successfully predicted anti-neutrophil cytoplasmic antibody−associated vasculitis with AUC >0.9 from clinical records (Wang et al, 2025).

Phenotyping, subtyping, and treatment personalization. A few studies that leverage AI to advance disease understanding exist. Hu et al (2024) phenotyped single-cell RNA and VDJ sequences on skin lesions from patients with bullous pemphigoid (BP), pemphigus vulgaris (PV), and controls, finding expanded CD8+ T-resident memory cells in PV and CD8+ T effector memory cells and CD4+ T regulatory cells in BP. Fortino et al (2020) phenotyped allergic contact dermatitis (ACD) and irritant contact dermatitis (ICD) with skin biopsy microarray gene expression modules, identifying biomarker genes such as *ADAM8* and *CD47* important in ACD and *RRM2* important in ICD. Bettuzzi et al (2022) utilized factor analysis and hierarchical clustering on baseline clinical and biological data for 113 patients with epidermal necrolysis, finding 3 clusters: 1 cluster of sicker patients with high mortality and 2 clusters of younger patients.

AI as a clinical aide for physicians and patients. Limited AI applications exist.

Summary. Isolated studies aim to leverage AI to classify inflammatory dermatoses or advance disease understanding with immunophenotyping. Newer AI models, including deep learning and generative AI applications, are underutilized (Omar et al, 2025).

CHALLENGES, OPPORTUNITIES, AND FUTURE DIRECTIONS

AI is rapidly transforming dermatology by providing novel tools to address the complex diagnostic and therapeutic challenges associated with skin diseases. Although early AI tools were predominantly focused on skin cancer detection, more recent advances in AI have expanded considerably to focus on inflammatory and autoimmune skin diseases. In this review, we discussed applications of AI across the spectrum of basic, translational, and clinical care, including disease classification/diagnosis, phenotyping, subtyping, and patient education. We outlined in depth disease-specific advances driven by AI/ML in AD, psoriasis vulgaris, HS, vitiligo, AA, and rheumatic skin diseases. However, there remains significant gaps and unmet needs in the field.

The predominant application of AI for inflammatory skin disease lies in disease classification and diagnosis because the first models were primarily trained on imaging data. With the emergence of large-scale multiomics and clinical data, subsequent studies within the last ∼5 years have now shifted to center on disease phenotyping, subtyping, precision medicine approaches, and drug discovery/repurposing. Fewest studies exist on generative AI applications in inflammatory skin disease, likely owing to the more recent development of transformer-based models. The primary focus of most of the AI-based studies are on AD and psoriasis, which is expected because they are 2 of the most common inflammatory skin diseases. Significantly fewer high-quality studies exist for acne, rosacea, AA, vitiligo, and HS. The fewest studies focused on rheumatologic skin disease (eg, lupus, DM, morphea/scleroderma, vasculitis) and autoimmune blistering disorders. Therefore, much more work is needed in the areas of AI applications for phenotyping, subtyping, and drug discovery for HS, vitiligo, and rheumatologic skin diseases because limited FDA-approved therapies exist. Given large amounts of available multiomics and clinical data, this represents an opportunity for AI-based computational drug repurposing and biomarker discovery.

The emergence of multimodal data integration, ranging from clinical images and histopathology to transcriptomic and genomic profiles, has enabled more refined

classification, disease subphenotyping, and identification of novel biomarkers and therapeutic targets (Figure 1). Looking ahead, the convergence of multimodal data analysis offers unprecedented opportunities for precision medicine in dermatology. By moving beyond images and integrating data such as genetic profiles, transcriptomics, exposomics, clinical information, and patient outcomes, AI tools can be utilized to match patients with optimal therapies on the basis of individual disease pathways and subphenotypes (Patrick et al, 2018a; Wongvibulsin et al, 2022). This approach could significantly reduce the trial-and-error nature of current treatment selection for possibly poorly defined diagnoses of inflammatory skin conditions. There is also an unmet need and opportunity for approaches to distinguish inflammatory conditions from infectious and malignant processes, particularly in cases where clinical presentations may overlap (Ujiie et al, 2022). In addition, AI-powered analysis of population-level data could identify communities at elevated risk for developing inflammatory conditions, enabling earlier interventions and preventive strategies.

The choice of AI paradigm and approach is influenced by the specific application and data available. Supervised learning remains the approach for well-defined classification tasks, such as diagnosis, where the label is determined to be of good quality. Trained models can provide value to non-dermatologists or clinicians in primary care settings where there may be less familiarity with inflammatory dermatoses. The supervised learning paradigm may be difficult if ground truth is not certain, such as in the unclear clinical diagnosis for some rheumatologic conditions. In those cases, various AI paradigms are utilized, including unsupervised learning approaches, with the aim to understand molecular endotypes or help with discovery-oriented tasks for researchers. Recent advances in transformers and large-scale foundation models may enable more complex applications, such as generating a differential diagnosis or summarizing patient history for clinicians or providing answers to patient questions, although these complex tasks present challenges with greater use of computational resources, lack of interpretability, and lack of ground truth for quality evaluation, which will require extensive monitoring and optimization before full clinical adoption.

The integration of generative AI and ML in dermatology presents both significant opportunities and important challenges for basic scientists and clinicians. Several key challenges persist. AI applications in inflammatory skin diseases beyond psoriasis and AD remain relatively understudied. Algorithmic bias, particularly related to the underrepresentation of skin of color, continues to limit the generalizability of AI models. Although some efforts aim to leverage generative AI to simulate various skin tones (Munia and Al Zubaer Imran, 2025[6]), these approaches also risk exacerbating health inequities and magnifying institutional racism (Groh et al, 2024; Grzybowski et al, 2024; Mikołajczyk et al, 2022[7]). A few studies propose methods to improve fairness across groups, such as model pruning (Wu et al, 2022b[8]). Misinformation is also rampant, with significant bias, inconsistent validation, and inadequate privacy protection (Wongvibulsin et al, 2024). Key technical hurdles include standardizing image quality across photo acquisition methods, addressing confounding factors and artifacts in images, and developing representative datasets that span diverse patient populations (Choy et al, 2023; Daneshjou et al, 2022), although preliminary models are evolving that can provide feedback on image quality during acquisition (Vodrahalli et al, 2023). Related to this, there has been significant discussion in the community regarding the potential harms of AI-based models, especially with respect to how hallucinations can lead to treatment recommendations and patient instructions that are factually incorrect or potentially detrimental/dangerous to both patients and the health system. In addition, the clinical translation of many AI tools remains in its infancy owing to limitations in data quality, external validation, and integration into existing workflows. Successful clinical implementation requires careful attention to workflow integration, provider education, and patient acceptance, ensuring that AI tools can enhance care delivery worldwide.

The future role of AI in dermatology will eventually include the integration of generative AI tools that support both routine clinical care and complex decision making. Generative AI models will enhance medical education, clinical simulation, and patient communication (Motolese et al, 2022; Zhang et al, 2024b), whereas improved analysis of immune mechanisms will enable the development of targeted treatments (Martínez et al, 2022; Seremet et al, 2024). Advancing the management of inflammatory skin disorders will require sustained collaboration among clinicians, researchers, pharmaceutical experts, and technology developers (Sengupta, 2023), with a strong focus on ethical considerations and equitable access to AI-enhanced care.

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CONFLICT OF INTEREST
The authors state no conflict of interest.

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[6] Munia N, Al Zubaer Imran A. Prompting medical vision-language models to mitigate diagnosis bias by generating realistic dermoscopic images. arXiv 2025.
[7] Mikołajczyk A, Majchrowska S, Limeros SC. The (de)biasing effect of GAN-based augmentation methods on skin lesion images. arXiv 2022.
[8] Wu Y, Zeng D, Xu X, Shi Y, Hu J. FairPrune: achieving fairness through pruning for dermatological disease diagnosis. arXiv 2022b

SUPPLEMENTARY MATERIAL

Supplementary material is linked to the online version of the paper at www.jidonline.org, and at 10.1016/j.jid.2025.10.596.

REFERENCES

Abhishek K, Kawahara J, Hamarneh G. Predicting the clinical management of skin lesions using deep learning. Sci Rep 2021;11:7769.

Absci. ABS-201. https://www.absci.com/abs-201-case-study/; 2025 (accessed June 16, 2025).

Aijaz SF, Khan SJ, Azim F, Shakeel CS, Hassan U. Deep learning application for effective classification of different types of psoriasis. J Healthc Eng 2022;2022:7541583.

Ali W, Williams J, Xiong B, Zou J, Daneshjou R. Machine learning for early detection of hidradenitis suppurativa: a feasibility study using medical insurance claims data. JID Innov 2025;5:100362.

Andreoletti G, Lanata CM, Trupin L, Paranjpe I, Jain TS, Nititham J, et al. Transcriptomic analysis of immune cells in a multi-ethnic cohort of systemic lupus erythematosus patients identifies ethnicity- and disease-specific expression signatures. Commun Biol 2021;4:488.

Aydin S, Karabacak M, Vlachos V, Margetis K. Large language models in patient education: a scoping review of applications in medicine. Front Med (Lausanne) 2024;11:1477898.

Barraza GA, Castro-Guijarro AC, De La Fuente Hoffmann V, Bolívar Ávila SJ, Flamini MI, Sanchez AM. Drug repositioning for rosacea disease: biological TARGET identification, molecular docking, pharmacophore mapping, and molecular dynamics analysis. Comput Biol Med 2024;181:108988.

Berna R, Mitra N, Hoffstad O, Wan J, Margolis DJ. Identifying phenotypes of atopic dermatitis in a longitudinal United States cohort using unbiased statistical clustering. J Invest Dermatol 2020;140:477–9.

Bettuzzi T, Hua C, Diaz E, Colin A, Wolkenstein P, De Prost N, et al. Epidermal necrolysis: characterization of different phenotypes using an unsupervised clustering analysis. Br J Dermatol 2022;186:1037–9.

Binol H, Plotner A, Sopkovich J, Kaffenberger B, Niazi MKK, Gurcan MN. Ros-NET: a deep convolutional neural network for automatic identification of rosacea lesions. Skin Res Technol 2020;26:413–21.

Capurro N, Pastore VP, Touijer L, Odone F, Cozzani E, Gasparini G, et al. A deep learning approach to direct immunofluorescence pattern recognition in autoimmune bullous diseases. Br J Dermatol 2024;191:261–6.

Cazzaniga S, Pezzolo E, Garcovich S, Naldi L, IRHIS Study Group. Late-onset hidradenitis suppurativa: a cluster analysis of the National Italian Registry IRHIS. J Am Acad Dermatol 2021;85:e29–32.

Cazzato G, Rongioletti F. Artificial intelligence in dermatopathology: updates, strengths, and challenges. Clin Dermatol 2024;42:437–42.

Ceccarelli F, Sciandrone M, Perricone C, Galvan G, Morelli F, Vicente LN, et al. Prediction of chronic damage in systemic lupus erythematosus by using machine-learning models. PLoS One 2017;12:e0174200.

Chakraborty C, Achar U, Nayek S, Achar A, Mukherjee R. CAD-PsorNet: deep transfer learning for computer-assisted diagnosis of skin psoriasis. Sci Rep 2024;14:26557.

Chen JC, Dai Z, Christiano AM. Regulatory network analysis defines unique drug mechanisms of action and facilitates patient-drug matching in alopecia areata clinical trials. Comput Struct Biotechnol J 2021;19:4751–8.

Choi S, Kim J, Lee JH, Lee YN, Lee JH. Clinical and histological characteristics of localized morphea, generalized morphea and systemic sclerosis: a comparative study aided by machine learning. Acta Derm Venereol 2023;103:adv11953.

Choy SP, Kim BJ, Paolino A, Tan WR, Lim SML, Seo J, et al. Systematic review of deep learning image analyses for the diagnosis and monitoring of skin disease. npj Digit Med 2023;6:180.

Clayton K, Vallejo A, Sirvent S, Davies J, Porter G, Reading IC, et al. Machine learning applied to atopic dermatitis transcriptome reveals distinct therapy-dependent modification of the keratinocyte immunophenotype. Br J Dermatol 2021;184:913–22.

Corbin A, Marques O. Assessing bias in skin lesion classifiers with contemporary deep learning and post-hoc explainability techniques. IEEE Access 2023;11:78339–52.

Cutts Z, Patterson S, Maliskova L, Taylor KE, Ye CJ, Dall'Era M, et al. Cell-specific transposable element and gene expression analysis across systemic lupus erythematosus phenotypes. ACR Open Rheumatol 2024;6:769–79.

Czajkowska J, Badura P, Korzekwa S, Płatkowska-Szczerek A. Deep learning approach to skin layers segmentation in inflammatory dermatoses. Ultrasonics 2021;114:106412.

Czajkowska J, Badura P, Korzekwa S, Płatkowska-Szczerek A. Automated segmentation of epidermis in high-frequency ultrasound of pathological skin using a cascade of DeepLab v3+ networks and fuzzy connectedness. Comput Med Imaging Graph 2022;95:102023.

Damiani G, Conic RRZ, Pigatto PDM, Carrera CG, Franchi C, Cattaneo A, et al. Predicting secukinumab fast-responder profile in psoriatic patients: advanced application of artificial-neural-networks (ANNs). J Drugs Dermatol 2020;19:1241–6.

Daneshjou R, Vodrahalli K, Novoa RA, Jenkins M, Liang W, Rotemberg V, et al. Disparities in dermatology AI performance on a diverse, curated clinical image set. Sci Adv 2022;8:eabq6147.

Deng Z, Chen M, Zhao Z, Xiao W, Liu T, Peng Q, et al. Whole genome sequencing identifies genetic variants associated with neurogenic inflammation in rosacea. Nat Commun 2023;14:3958.

Dev K, Ho CJH, Bi R, Yew YW, DU S, Attia ABE, et al. Machine learning assisted handheld confocal Raman micro-spectroscopy for identification of clinically relevant atopic eczema biomarkers. Sensors (Basel) 2022;22:4674.

Esteva A, Kuprel B, Novoa RA, Ko J, Swetter SM, Blau HM, et al. Dermatologist-level classification of skin cancer with deep neural networks [published correction appears in: Nature 2017;546:686]. Nature 2017;542:115–8.

Félix Garza ZC, Lenz M, Liebmann J, Ertaylan G, Born M, Arts ICW, et al. Characterization of disease-specific cellular abundance profiles of chronic inflammatory skin conditions from deconvolution of biopsy samples. BMC Med Genomics 2019;12:121.

Fortino V, Wisgrill L, Werner P, Suomela S, Linder N, Jalonen E, et al. Machine-learning-driven biomarker discovery for the discrimination between allergic and irritant contact dermatitis. Proc Natl Acad Sci USA 2020;117:33474–85.

Franks JM, Martyanov V, Cai G, Wang Y, Li Z, Wood TA, et al. A machine learning classifier for assigning individual patients with systemic sclerosis to intrinsic molecular subsets. Arthritis Rheumatol 2019;71:1701–10.

Gao Y, Xu C, Yan Q. Genetic analyses reveal association between rosacea and cardiovascular diseases. Sci Rep 2025;15:6657.

Ge L, Li Y, Wu Y, Fan Z, Song Z. Differential diagnosis of rosacea using machine learning and dermoscopy. Clin Cosmet Investig Dermatol 2022;15:1465–73.

Ghosh D, Ding L, Sivaprasad U, Geh E, Biagini Myers J, Bernstein JA, et al. Multiple transcriptome data analysis reveals biologically relevant atopic dermatitis signature genes and pathways. PLoS One 2015;10:e0144316.

González-Manso A, Agut-Busquet E, Romaní J, Vilarrasa E, Bittencourt F, Mensa A, et al. Hidradenitis suppurativa: proposal of classification in two endotypes with two-step cluster analysis. Dermatology 2021;237:365–71.

Greenfield DA, Feizpour A, Evans CL. Quantifying inflammatory response and drug-aided resolution in an atopic dermatitis model with deep learning. J Invest Dermatol 2023;143:1430–8.e4.

Groh M, Badri O, Daneshjou R, Koochek A, Harris C, Soenksen LR, et al. Deep learning-aided decision support for diagnosis of skin disease across skin tones. Nat Med 2024;30:573–83.

Grzybowski A, Jin K, Wu H. Challenges of artificial intelligence in medicine and dermatology. Clin Dermatol 2024;42:210–5.

Guo L, Yang Y, Ding H, Zheng H, Yang H, Xie J, et al. A deep learning-based hybrid artificial intelligence model for the detection and severity assessment of vitiligo lesions. Ann Transl Med 2022;10:590.

Gustafson E, Pacheco J, Wehbe F, Silverberg J, Thompson W. A machine learning algorithm for identifying atopic dermatitis in adults from electronic health records. Proc (IEEE Int Conf Healthc Inform) 2017;2017:83–90.

Hammad M, Pławiak P, ElAffendi M, El-Latif AAA, Latif AAA. Enhanced deep learning approach for accurate eczema and psoriasis skin detection. Sensors 2023;23:7295.

Hawkes JE, Chan TC, Krueger JG. Psoriasis pathogenesis and the development of novel targeted immune therapies. J Allergy Clin Immunol 2017;140:645–53.

He H, Bissonnette R, Wu J, Diaz A, Saint-Cyr Proulx E, Maari C, et al. Tape strips detect distinct immune and barrier profiles in atopic dermatitis and psoriasis. J Allergy Clin Immunol 2021;147:199–212.

Hillmer D, Merhi R, Boniface K, Taieb A, Barnetche T, Seneschal J, et al. Evaluation of facial vitiligo severity with a mixed clinical and artificial intelligence approach. J Invest Dermatol 2024;144:351–7.e4.

Ho CJH, Yew YW, Dinish US, Kuan AHY, Wong MKW, Bi R, et al. Handheld confocal Raman spectroscopy (CRS) for objective assessment of skin barrier function and stratification of severity in atopic dermatitis (AD) patients. J Dermatol Sci 2020;98:20–5.

Hosny KM, Kassem MA, Fouad MM. Classification of skin lesions into seven classes using transfer learning with AlexNet. J Digit Imaging 2020;33:1325–34.

Hu Z, Zheng M, Guo Z, Zhou W, Zhou W, Yao N, et al. Single-cell sequencing reveals distinct immune cell features in cutaneous lesions of pemphigus vulgaris and bullous pemphigoid. Clin Immunol 2024;263:110219.

Hua VJ, Kilgour JM, Cho HG, Li S, Sarin KY. Characterization of comorbidity heterogeneity among 13,667 patients with hidradenitis suppurativa. JCI Insight 2021;6:e151872.

Huang K, Jiang Z, Li Y, Wu Z, Wu X, Zhu W, et al. The classification of six common skin diseases based on Xiangya-derm: development of a Chinese database for artificial intelligence. J Med Internet Res 2021a;23:e26025.

Huang K, Wu X, Li Y, Lv C, Yan Y, Wu Z, et al. Artificial intelligence-based psoriasis severity assessment: real-world study and application. J Med Internet Res 2023;25:e44932.

Huang Y, Wen HJ, Guo YL, Wei TY, Wang WC, Tsai SF, et al. Prenatal exposure to air pollutants and childhood atopic dermatitis and allergic rhinitis adopting machine learning approaches: 14-year follow-up birth cohort study. Sci Total Environ 2021b;777:145982.

Hurault G, Domínguez-Hüttinger E, Langan SM, Williams HC, Tanaka RJ. Personalized prediction of daily eczema severity scores using a mechanistic machine learning model. Clin Exp Allergy 2020;50:1258–66.

Jamian L, Wheless L, Crofford LJ, Barnado A. Rule-based and machine learning algorithms identify patients with systemic sclerosis accurately in the electronic health record. Arthritis Res Ther 2019;21:305.

Jiang Z, Li J, Kong N, Kim JH, Kim BS, Lee MJ, et al. Accurate diagnosis of atopic dermatitis by combining transcriptome and microbiota data with supervised machine learning. Sci Rep 2022;12:290.

Johri S, Jeong J, Tran BA, Schlessinger DI, Wongvibulsin S, Barnes LA, et al. An evaluation framework for clinical use of large language models in patient interaction tasks. Nat Med 2025;31:77–86.

Kardynska M, Kogut D, Pacholczyk M, Smieja J. Mathematical modeling of regulatory networks of intracellular processes - Aims and selected methods. Comput Struct Biotechnol J 2023;21:1523–32.

Karthik R, Vaichole TS, Kulkarni SK, Yadav O, Khan F. Eff2Net: an efficient channel attention-based convolutional neural network for skin disease classification. Biomed Signal Process Control 2022;73:103406.

Khan Y, Todorov A, Torah R, Beeby S, Ardern-Jones MR. Skin sensing and wearable technology as tools to measure atopic dermatitis severity. Skin Health Dis 2024;4:e449.

Khera R, Butte AJ, Berkwits M, Hswen Y, Flanagin A, Park H, et al. AI in medicine-JAMA's focus on clinical outcomes, patient-centered care, quality, and equity. JAMA 2023;330:818–20.

Kim C, Gadgil SU, DeGrave AJ, Omiye JA, Cai ZR, Daneshjou R, et al. Transparent medical image AI via an image-text foundation model grounded in medical literature. Nat Med 2024;30:1154–65.

Kirby J, Kim K, Zivkovic M, Wang S, Garg V, Danavar A, et al. Uncovering the burden of hidradenitis suppurativa misdiagnosis and underdiagnosis: a machine learning approach. Front Med Technol 2024;6:1200400.

Koopman JJE, Buhler KA, Choi MY. From machine learning to clinical practice: phenotypic clusters of anti-MDA5 antibody-positive dermatomyositis. Brief Bioinform 2024;25:bbae303.

Lakdawala N, Channa L, Gronbeck C, Lakdawala N, Weston G, Sloan B, et al. Assessing the accuracy and comprehensiveness of ChatGPT in offering clinical guidance for atopic dermatitis and acne vulgaris. JMIR Dermatol 2023;6:e50409.

Lee S, Lee JW, Choe SJ, Yang S, Koh SB, Ahn YS, et al. Clinically applicable deep learning framework for measurement of the extent of hair loss in patients with alopecia areata. JAMA Dermatol 2020;156:1018–20.

Li Pomi F, Papa V, Borgia F, Vaccaro M, Pioggia G, Gangemi S. Artificial intelligence: a snapshot of its application in chronic inflammatory and autoimmune skin diseases. Life (Basel) 2024;14:516.

Li Q, Yang Z, Chen K, Zhao M, Long H, Deng Y, et al. Human-multimodal deep learning collaboration in 'precise' diagnosis of lupus erythematosus subtypes and similar skin diseases. J Eur Acad Dermatol Venereol 2024;38:2268–79.

Li X, Zhao X, Ma H, Xie B. Image analysis and diagnosis of skin diseases - a review. Curr Med Imaging Rev 2023b;19:199–242.

Li Y, Li L, Tian Y, Luo J, Huang J, Zhang L, et al. Identification of novel immune subtypes and potential hub genes of patients with psoriasis. J Transl Med 2023a;21:182.

Li Y, Ma C, Liao S, Qi S, Meng S, Cai W, et al. Combined proteomics and single cell RNA-sequencing analysis to identify biomarkers of disease diagnosis and disease exacerbation for systemic lupus erythematosus. Front Immunol 2022;13:969509.

Lin B, Xu Y, Bao X, Zhao Z, Wang Z, Yin J. SkinGEN: an explainable dermatology diagnosis-to-generation framework with interactive vision-language models. Cagliari, Italy: Paper presented at: IUI '25: 30th International Conference on Intelligent User Interfaces; 24–27 March 2025.

Liu S, Fan Y, Duan M, Wang Y, Su G, Ren Y, et al. AcneGrader: an ensemble pruning of the deep learning base models to grade acne. Skin Res Technol 2022a;28:677–88.

Liu Y, Jain A, Eng C, Way DH, Lee K, Bui P, et al. A deep learning system for differential diagnosis of skin diseases. Nat Med 2020;26:900–8.

Liu Y, Wang H, Taylor M, Cook C, Martínez-Berdeja A, North JP, et al. Classification of human chronic inflammatory skin disease based on single-cell immune profiling. Sci Immunol 2022b;7:eabl9165.

Klçoğlu MS, Yurdakul O, Aydn T. Evaluation of chat generative pretrained transformer's responses to frequently asked questions about psoriatic arthritis: a study on quality and readability. Ann Med Res 2025;32:78–83.

Ma W, Lau YL, Yang W, Wang YF. Random forests algorithm boosts genetic risk prediction of systemic lupus erythematosus. Front Genet 2022;13:902793.

Maintz L, Welchowski T, Herrmann N, Brauer J, Kläschen AS, Fimmers R, et al. Machine learning–based deep phenotyping of atopic dermatitis: severity-associated factors in adolescent and adult patients. JAMA Dermatol 2021;157:1414–24.

Mao R, Li J. Construction of a molecular diagnostic system for neurogenic rosacea by combining transcriptome sequencing and machine learning. BMC Med Genomics 2024;17:232.

Martínez BA, Shrotri S, Kingsmore KM, Bachali P, Grammer AC, Lipsky PE. Machine learning reveals distinct gene signature profiles in lesional and nonlesional regions of inflammatory skin diseases. Sci Adv 2022;8:eabn4776.

McDuff D, Schaekermann M, Tu T, Palepu A, Wang A, Garrison J, et al. Towards accurate differential diagnosis with large language models. Nature 2025;642:451–7.

Mendel T, Singh N, Mann DM, Wiesenfeld B, Nov O. Laypeople's use of and attitudes toward large language models and search engines for health queries: survey study. J Med Internet Res 2025;27:e64290.

Miao BY, Chen IY, Williams CYK, Davidson J, Garcia-Agundez A, Sun S, et al. The MI-CLAIM-GEN checklist for generative artificial intelligence in health. Nat Med 2025;31:1394–8.

Mohan J, Sivasubramanian A, V S, Ravi V. Enhancing skin disease classification leveraging transformer-based deep learning architectures and explainable AI. Comput Biol Med 2025;190:110007.

Moon CI, Kim EB, Baek YS, Lee O. Transformer based on the prediction of psoriasis severity treatment response. Biomed Signal Process Control 2024;89:105743.

Motolese A, Ceccarelli M, Macca L, Li Pomi F, Ingrasciotta Y, Nunnari G, et al. Novel therapeutic approaches to psoriasis and risk of infectious disease. Biomedicines 2022;10:228.

Muhaba KA, Dese K, Aga TM, Zewdu FT, Simegn GL. Automatic skin disease diagnosis using deep learning from clinical image and patient information. Skin Health Dis 2022;2:e81.

Muñoz-López C, Ramírez-Cornejo C, Marchetti MA, Han SS, Del Barrio-Díaz P, Jaque A, et al. Performance of a deep neural network in teledermatology: a single-centre prospective diagnostic study. J Eur Acad Dermatol Venereol 2021;35:546–53.

Nair S, Lushington GH, Purushothaman M, Rubin B, Jupe E, Gattam S. Prediction of lupus classification criteria via generative AI medical record profiling. BioTech (Basel) 2025;14:15.

Neely J, Hartoularos G, Bunis D, Sun Y, Lee D, Kim S, et al. Multi-modal single-cell sequencing identifies cellular immunophenotypes associated with juvenile dermatomyositis disease activity. Front Immunol 2022;13:902232.

Nicholas A, Spraul A, Fleischer AB Jr. Prescriber phenotypes: variability in topical rosacea treatment patterns among United States dermatologists. J Clin Med 2024;13:6275.

Okamoto T, Kawai M, Ogawa Y, Shimada S, Kawamura T. Artificial intelligence for the automated single-shot assessment of psoriasis severity. J Eur Acad Dermatol Venereol 2022;36:2512–5.

Omar M, Agbareia R, Naffaa ME, Watad A, Glicksberg BS, Nadkarni GN, et al. Applications of artificial intelligence in vasculitides: a systematic review. ACR Open Rheumatol 2025;7:e70016.

Papa V, Li Pomi F, Borgia F, Genovese S, Pioggia G, Gangemi S. "Mens Sana in Cute Sana"-a state of the art of mutual etiopathogenetic influence and relevant pathophysiological pathways between skin and mental disorders: an integrated approach to contemporary psychopathological scenarios. Cells 2023;12:1828.

Park S, Chien AL, Lin B, Li K. FACES: a deep-learning-based parametric model to improve rosacea diagnoses. Appl Sci (Basel) 2023;13:970.

Patrick MT, Raja K, Miller K, Sotzen J, Gudjonsson JE, Elder JT, et al. Drug repurposing prediction for immune-mediated cutaneous diseases using a word-embedding-based machine learning approach. J Invest Dermatol 2018a;139:683–91.

Patrick MT, Stuart PE, Raja K, Gudjonsson JE, Tejasvi T, Yang J, et al. Genetic signature to provide robust risk assessment of psoriatic arthritis development in psoriasis patients. Nat Commun 2018b;9:4178.

Pournara E, Kormaksson M, Nash P, Ritchlin CT, Kirkham BW, Ligozio G, et al. Clinically relevant patient clusters identified by machine learning from the clinical development programme of secukinumab in psoriatic arthritis. RMD Open 2021;7:e001845.

Prakash AV, Park JW, Seong JW, Kang TJ. Repositioned Drugs for inflammatory Diseases such as Sepsis, Asthma, and Atopic Dermatitis. Biomol Ther (Seoul) 2020;28:222–9.

Prasannanjaneyulu V, Nene S, Jain H, Nooreen R, Otavi S, Chitlangya P, et al. Old drugs, new tricks: emerging role of drug repurposing in the management of atopic dermatitis. Cytokine Growth Factor Rev 2022;65:12–26.

Queiro R, Seoane-Mato D, Laiz A, Agirregoikoa EG, Montilla C, Park HS, et al. Minimal disease activity (MDA) in patients with recent-onset psoriatic arthritis: predictive model based on machine learning. Arthritis Res Ther 2022;24:153.

Rajalingam K, Levin N, Marques O, Grichnik J, Lin A, Chen WS. Treatment options and emotional well-being in patients with rosacea: an unsupervised machine learning analysis of over 200,000 posts. JAAD Int 2023;13:172–8.

Schaap MJ, Cardozo NJ, Patel A, De Jong EMGJ, Van Ginneken B, Seyger MMB. Image-based automated Psoriasis Area Severity Index scoring by Convolutional Neural Networks. J Eur Acad Dermatol Venereol 2022;36:68–75.

Sengupta D. Artificial intelligence in diagnostic dermatology: challenges and the Way forward. Indian Dermatol Online J 2023;14:782–7.

Seremet T, Di Domizio J, Girardin A, Yatim A, Jenelten R, Messina F, et al. Immune modules to guide diagnosis and personalized treatment of inflammatory skin diseases. Nat Commun 2024;15:10688.

Shapiro J, Lyakhovitsky A. Revolutionizing teledermatology: exploring the integration of artificial intelligence, including Generative Pre-trained Transformer chatbots for artificial intelligence-driven anamnesis, diagnosis, and treatment plans. Clin Dermatol 2024;42:492–7.

Shrivastava VK, Londhe ND, Sonawane RS, Suri JS. A novel approach to multiclass psoriasis disease risk stratification: machine learning paradigm. Biomed Signal Process Control 2016;28:27–40.

Shrivastava VK, Londhe ND, Sonawane RS, Suri JS. A novel and robust Bayesian approach for segmentation of psoriasis lesions and its risk stratification. Comput Methods Programs Biomed 2017;150:9–22.

Soenksen LR, Kassis T, Conover ST, Marti-Fuster B, Birkenfeld JS, Tucker-Schwartz J, et al. Using deep learning for dermatologist-level detection of suspicious pigmented skin lesions from wide-field images. Sci Transl Med 2021;13:eabb3652.

Sulejmani P, Negris O, Aoki V, Chu CY, Eichenfield L, Misery L, et al. A large language model artificial intelligence for patient queries in atopic dermatitis. J Eur Acad Dermatol Venereol 2024;38:e531–5.

Tang AS, Rankin KP, Cerono G, Miramontes S, Mills H, Roger J, et al. Leveraging electronic health records and knowledge networks for Alzheimer's disease prediction and sex-specific biological insights. Nat Aging 2024;4:379–95.

Tao Y, Hua G, Min S, Xiao-Hong L, Jia-Hui J, Biao T, et al. Verification of biological markers of subacute cutaneous lupus erythematosus via TMT labelling proteomics combined with transcriptome data. Ann Med 2025;57:2500696.

Tomalin LE, Kim J, Correa Da Rosa J, Lee J, Fitz LJ, Berstein G, et al. Early quantification of systemic inflammatory proteins predicts long-term treatment response to tofacitinib and etanercept. J Invest Dermatol 2020;140:1026–34.

Tu T, Schaekermann M, Palepu A, Saab K, Freyberg J, Tanno R, et al. Towards conversational diagnostic artificial intelligence. Nature 2025;642:442–50.

Ujiie H, Rosmarin D, Schön MP, Ständer S, Boch K, Metz M, et al. Unmet medical needs in chronic, non-communicable inflammatory skin diseases. Front Med (Lausanne) 2022;9:875492.

van der Schaft J, Thijs JL, Garritsen FM, Balak D, de Bruin-Weller MS. Towards personalized treatment in atopic dermatitis. Expert Opin Biol Ther 2019;19:469–76.

Vodrahalli K, Daneshjou R, Novoa RA, Chiou A, Ko JM, Zou J. TrueImage: a machine learning algorithm to improve the quality of telehealth photos. Pac Symp Biocomput 2021;26:220–31.

Vodrahalli K, Ko J, Chiou AS, Novoa R, Abid A, Phung M, et al. Development and clinical evaluation of an artificial intelligence support tool for improving telemedicine photo quality. JAMA Dermatol 2023;159:496–503.

Wang A, Fulton R, Hwang S, Margolis DJ, Mowery D. Patient phenotyping for atopic dermatitis with transformers and machine learning: algorithm development and validation study. JMIR Form Res 2024b;8:e52200.

Wang D, Pang N, Wang Y, Zhao H. Unlabeled skin lesion classification by self-supervised topology clustering network. Biomed Signal Process Control 2021b;66:102428.

Wang J, Luo L, Ding Q, Wu Z, Peng Y, Li J, et al. Development of a multi-target strategy for the treatment of vitiligo via machine learning and network analysis methods. Front Pharmacol 2021a;12:754175.

Wang L, Novoa-Laurentiev J, Cook C, Srivatsan S, Hua Y, Yang J, et al. Identification of an ANCA-associated vasculitis cohort using deep learning and electronic health records. Int J Med Inform 2025;196:105797.

Wang X, Hu H, Yan G, Zheng B, Luo J, Fan J. Identification and validation of interferon-stimulated gene 15 as a biomarker for dermatomyositis by integrated bioinformatics analysis and machine learning. Front Immunol 2024c;15:1429817.

Wang Y, Chen J, Shen ZY, Zhang J, Zhu YJ, Xia XQ. Screening of diagnostic biomarkers and immune infiltration characteristics linking rheumatoid arthritis and rosacea based on bioinformatics analysis. J Inflamm Res 2024a;17:5177–95.

Wang Y, Qin D, Jin L, Liang G. Caffeoyl malic acid is a potential dual inhibitor targeting TNFα/IL-4 evaluated by a combination strategy of network analysis-deep learning-molecular simulation. Comput Biol Med 2022;145:105410.

Wiala A, Ranjan R, Schnidar H, Rappersberger K, Posch C. Automated classification of hidradenitis suppurativa disease severity by convolutional neural network analyses using calibrated clinical images. J Eur Acad Dermatol Venereol 2024;38:576–82.

Williams CYK, Miao BY, Kornblith AE, Butte AJ. Evaluating the use of large language models to provide clinical recommendations in the Emergency Department. Nat Commun 2024;15:8236.

Wongvibulsin S, Frech TM, Chren MM, Tkaczyk ER. Expanding personalized, data-driven dermatology: leveraging digital health technology and machine learning to improve patient outcomes. JID Innov 2022;2:100105.

Wongvibulsin S, Yan MJ, Pahalyants V, Murphy W, Daneshjou R, Rotemberg V. Current state of dermatology mobile applications with artificial intelligence features [published correction appears in JAMA Dermatol 2024;160:688]. JAMA Dermatol 2024;160:646–50.

Wu H, Yin H, Chen H, Sun M, Liu X, Yu Y, et al. A deep learning-based smartphone platform for cutaneous lupus erythematosus classification assistance: simplifying the diagnosis of complicated diseases. J Am Acad Dermatol 2021;85:792–3.

Wu H, Yin H, Chen H, Sun M, Liu X, Yu Y, et al. A deep learning, image based approach for automated diagnosis for inflammatory skin diseases. Ann Transl Med 2020;8:581.

Wu JJ, Hong CH, Merola JF, Gruben D, Güler E, Feeney C, et al. Predictors of nonresponse to dupilumab in patients with atopic dermatitis: a machine learning analysis. Ann Allergy Asthma Immunol 2022a;129:354–9.e5.

Wu W, Chen G, Zhang Z, He M, Li H, Yan F. Construction and verification of atopic dermatitis diagnostic model based on pyroptosis related biological markers using machine learning methods. BMC Med Genomics 2023;16:138.

Xiong J, Chen G, Liu Z, Wu X, Xu S, Xiong J, et al. Construction of regulatory network for alopecia areata progression and identification of immune monitoring genes based on multiple machine-learning algorithms. Precis Clin Med 2023;6:pbad009.

Xue Y, Zhang J, Li C, Liu X, Kuang W, Deng J, et al. Machine learning for screening and predicting the risk of anti-MDA5 antibody in juvenile dermatomyositis children. Front Immunol 2023;13:940802.

Yamanaka C, Uki S, Kaitoh K, Iwata M, Yamanishi Y. De novo drug design based on patient gene expression profiles via deep learning. Mol Inform 2023;42:e2300064.

Yan S, Yu Z, Primiero C, Vico-Alonso C, Wang Z, Yang L, et al. A multimodal vision foundation model for clinical dermatology. Nat Med 2025;31:2691–702.

Yang AF, Patel S, Chun KS, Richards D, Walter JR, Okamoto K, et al. Artificial intelligence-enabled wearable devices and nocturnal scratching in mild atopic dermatitis. JAMA Dermatol 2025;161:406–10.

Yang Y, Guo L, Wu Q, Zhang M, Zeng R, Ding H, et al. Construction and evaluation of a deep learning model for assessing acne vulgaris using clinical images. Dermatol Ther (Heidelb) 2021;11:1239–48.

Yaung KN, Yeo JG, Kumar P, Wasser M, Chew M, Ravelli A, et al. Artificial intelligence and high-dimensional technologies in the theragnosis of systemic lupus erythematosus. Lancet Rheumatol 2023;5:e151–65.

Young AT, Fernandez K, Pfau J, Reddy R, Cao NA, von Franque MY, et al. Stress testing reveals gaps in clinic readiness of image-based diagnostic artificial intelligence models. NPJ Digit Med 2021;4:10.

Yu K, Syed MN, Bernardis E, Gelfand JM. Machine learning applications in the evaluation and management of psoriasis: a systematic review. J Psoriasis Psoriatic Arthritis 2020;5:147–59.

Yu Z, Kaizhi S, Jianwen H, Guanyu Y, Yonggang W. A deep learning-based approach toward differentiating scalp psoriasis and seborrheic dermatitis from dermoscopic images. Front Med (Lausanne) 2022;9:965423.

Zhang S, Chang M, Zheng L, Wang C, Zhao R, Song S, et al. Deep analysis of skin molecular heterogeneities and their significance on the precise treatment of patients with psoriasis. Front Immunol 2024a;15:1326502.

Zhang T, Nie Y. Prediction of the risk of alopecia areata progressing to alopecia totalis and alopecia universalis: biomarker development with bioinformatics analysis and machine learning. Dermatology 2022;238:386–96.

Zhang Y, Chen R, Nguyen D, Choi S, Gabel C, Leonard N, et al. Assessing the ability of an artificial intelligence chatbot to translate dermatopathology reports into patient-friendly language: a cross-sectional study. J Am Acad Dermatol 2024b;90:397–9.

Zhao S, Xie B, Li Y, Zhao X, Kuang Y, Su J, et al. Smart identification of psoriasis by images using convolutional neural networks: a case study in China. J Eur Acad Dermatol Venereol 2020;34:518–24.

Zhao Z, Wu CM, Zhang S, He F, Liu F, Wang B, et al. A novel convolutional neural network for the diagnosis and classification of rosacea: usability study. JMIR Med Inform 2021;9:e23415.

Zhong F, He K, Ji M, Chen J, Gao T, Li S, et al. Optimizing vitiligo diagnosis with ResNet and Swin transformer deep learning models: a study on performance and interpretability. Sci Rep 2024;14:9127.

Zhou J, He X, Sun L, Xu J, Chen X, Chu Y, et al. Pre-trained multimodal large language model enhances dermatological diagnosis using SkinGPT-4. Nat Commun 2024a;15:5649.

Zhou X, Zhou H, Luo X, Wu RF. Discovery of biomarkers in the psoriasis through machine learning and dynamic immune infiltration in three types of skin lesions. Front Immunol 2024b;15:1388690.

Zhu CY, Wang YK, Chen HP, Gao KL, Shu C, Wang JC, et al. A deep learning based framework for diagnosing multiple skin diseases in a clinical environment. Front Med (Lausanne) 2021b;8:626369.

Zhu JL, Tran LT, Smith M, Zheng F, Cai L, James JA, et al. Modular gene analysis reveals distinct molecular signatures for subsets of patients with cutaneous lupus erythematosus. Br J Dermatol 2021a;185:563–72.

Zou J, Topol EJ. The rise of agentic AI teammates in medicine. Lancet 2025;405:457.

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SUPPLEMENTARY MATERIALS

Supplementary Table S1. Studies in AI-Enabled Advancements across the Spectrum of Inflammatory Skin Diseases

	Application		
Diseases	**Classification and Diagnosis**	**Phenotyping, Subtyping, and Treatment Personalization**	**AI for Workflows and Patient Care**
Atopic dermatitis	Dev et al, 2022; Fortino et al, 2020; Ghosh et al, 2015; Hammad et al, 2023; Ho et al, 2020; Huang et al, 2021b; Hurault et al, 2020; Jiang et al, 2022; Liu et al, 2020; Muhaba et al, 2022; Muñoz-López et al, 2021; Wu et al, 2020; Zhu et al, 2021b	Berna et al, 2020; Clayton et al, 2021; Dev et al, 2022; Fortino et al, 2020; Greenfield et al, 2023; Gustafson et al, 2017; Jiang et al, 2022; Khan et al, 2024; Maintz et al, 2021; Martínez et al, 2022; Prabhu et al, 2018[9]; Prakash et al, 2020; Wang et al, 2024b, 2022; Wu et al, 2023, 2022a	Czajkowska et al, 2022, 2021a; Fortino et al, 2020; Giavina Bianchi et al, 2024; Ho et al, 2020; Huang et al, 2021b; Khan et al, 2024; Lakdawala et al, 2023; Maintz et al, 2021; Prakash et al, 2020; Prasannanjaneyulu et al, 2022; Sulejmani et al, 2024; Yang et al, 2025; Zhang et al, 2024b
Psoriasis	Aijaz et al, 2022; Chakraborty et al, 2024; Eskandari and Sharbatdar, 2024; Hammad et al, 2023; Huang et al, 2021a; Hurault et al, 2020; Karthik et al, 2022; Liu et al, 2020; Shrivastava et al, 2016; Wu et al, 2020; Yu et al, 2022; Zhao et al, 2020; Zhou et al, 2024b; Zhu et al, 2021b	Aijaz et al, 2022; Damiani et al, 2020; Martínez et al, 2022; Moon et al, 2024; Okamoto et al, 2022; Patrick et al, 2018a, 2018b; Pournara et al, 2021; Shrivastava et al, 2017, 2016; Tomalin et al, 2020; Xu and Zhang, 2017; Zhang et al, 2024a; Zhou et al, 2024b	Czajkowska et al, 2022, 2021b; Klçoğlu et al., 2025; Lin et al, 2025; Moon et al, 2024; Okamoto et al, 2022; Queiro et al, 2022; Schaap et al, 2022; Tomalin et al, 2020; Xu and Zhang, 2017; Yu et al, 2020; Zhang et al, 2024a, 2024b
Rosacea	Ge et al, 2022; Park et al, 2023; Zhao et al, 2021; Zhu et al, 2021b	Barraza et al, 2024; Binol et al, 2020; Deng et al, 2023; Gao et al, 2025; Ge et al, 2022; Mao and Li, 2024; Nicholas et al, 2024; Rajalingam et al, 2023; T et al, 2025; Wang et al, 2024a	Barraza et al, 2024; Nicholas et al, 2024; Rajalingam et al, 2023
Acne	Karthik et al, 2022; Liu et al, 2020; Muhaba et al, 2022; Muñoz-López et al, 2021	Abdel Mawgoud and Posch, 2024; Liu et al, 2022a; Yang et al, 2021	Abdel Mawgoud and Posch, 2024; Lakdawala et al, 2023; Yang et al, 2021
Hidradenitis suppurativa	Ali et al, 2025; Kirby et al, 2024; Muñoz-López et al, 2021	Cazzaniga et al, 2021; González-Manso et al, 2021; Wiala et al, 2024	
Alopecia areata	Liu et al, 2020; Muñoz-López et al, 2021; Zhang and Nie, 2022	Absci, 2025; Chen et al, 2021; Xiong et al, 2023; Zhang and Nie, 2022	
Vitiligo	Guo et al, 2022; Zhong et al, 2024	Hillmer et al, 2024; Wang et al, 2021a	
Cutaneous lupus erythematosus	Li et al, 2024; Nair et al, 2025; Wu et al, 2021	Dunlap et al, 2022; Martínez et al, 2022; Perez et al, 2022; Tao et al, 2025; Zhu et al, 2021a	
Morphea or scleroderma	Huang et al, 2021a	Abdel Mawgoud and Posch, 2024; Choi et al, 2023; Franks et al, 2019; Martínez et al, 2022	Abdel Mawgoud and Posch, 2024
Dermatomyositis	Xue et al, 2023	Koopman et al, 2024; Neely et al, 2022; Wang et al, 2024c	
Other or multiple	3Derm Systems, Inc, 2020; Capurro et al, 2024; Ahmad et al, 2022; Corbin and Marques, 2023; Czajkowska et al, 2021b; Esteva et al, 2017; Huang et al, 2021a; Lin et al, 2025; Liu et al, 2022b, 2022c, 2020; McDuff et al, 2025; Mohan et al, 2025; Muhaba et al, 2022; Muñoz-López et al, 2021; Shen et al, 2024; Wang et al, 2021b; Wu et al, 2022b[10]; Yan et al, 2025a[11], 2025b; Zhang et al, 2024c; Zhou et al, 2024a; Zhu et al, 2021b	Abdel Mawgoud and Posch, 2024; Abhishek et al, 2021; Bettuzzi et al, 2022; Esteva et al, 2017; Hart et al, 2024; Hu et al, 2024; Koopman et al, 2024; Liu et al, 2022c, 2020; McLeish et al, 2023; Mikołajczyk et al, 2022[12]; Neely et al, 2022; Patrick et al, 2018a; Prabhu et al, 2018[1]; Seremet et al, 2024; Shanehsazzadeh et al, 2023[13]; Shen et al, 2024; Wang et al, 2024c, 2021b; Wu et al, 2020a; Xue et al, 2023; Ying et al, 2024[14]; and Zhu et al, 2021b	3Derm Systems, Inc, 2020; Abdel Mawgoud and Posch, 2024; Abhishek et al, 2021; Czajkowska et al, 2022, 2021a; Groh et al, 2024; Jütte et al, 2024; Kim et al, 2024; Koopman et al, 2024; Lin et al, 2025; Mikołajczyk et al, 2022[4]; Muhaba et al, 2022; Munia and Al Zubaer Imran, 2025[15]; Muñoz-López et al, 2021; Shanehsazzadeh et al, 2023[5]; Vodrahalli et al, 2021; Yan et al, 2025b; Zhang et al, 2024b, 2024c; Zhou et al, 2024a

Abbreviation: AI, artificial intelligence.

[9]Prabhu V, Kannan A, Ravuri M, Chablani M, Sontag D, Amatriain X. Prototypical clustering networks for dermatological disease diagnosis. arXiv 2018.

[10]Wu Y, Zeng D, Xu X, Shi Y, Hu J. FairPrune: achieving fairness through pruning for dermatological disease diagnosis. arXiv 2022b.

[11]Yan S, Li X, Hu M, Jiang Y, Yu Z, Ge Z. MAKE: multi-aspect knowledge-enhanced vision-language pretraining for zero-shot dermatological assessment. arXiv 2025a.

[12]Mikołajczyk A, Majchrowska S, Limeros SC. The (de)biasing effect of GAN-based augmentation methods on skin lesion images. arXiv 2022.

[13]Shanehsazzadeh A, McPartlon M, Kasun G, Steiger AK, Sutton JM, Yassine E, et al. Unlocking *de novo* antibody design with generative artificial intelligence. bioRxiv 2023.

[14]Ying W, Wang D, Hu X, Qiu J, Park J, Fu Y. Revolutionizing biomarker discovery: leveraging generative AI for bio-knowledge-embedded continuous space exploration. arXiv 2024.

[15]Munia N, Al Zubaer Imran A. Prompting medical vision-language models to mitigate diagnosis bias by generating realistic dermoscopic images arXiv 2025.

SUPPLEMENTARY REFERENCES

3Derm Systems, Inc. 3Derm announces two FDA breakthrough device designations for autonomous skin cancer AI. https://www.prnewswire.com/news-releases/3derm-announces-two-fda-breakthrough-device-designations-for-autonomous-skin-cancer-ai-300982072.html; 2020. (accessed June 16, 2025).

Abdel Mawgoud R, Posch C. Visual explainability of 250 skin diseases viewed through the eyes of an AI-based, self-supervised vision transformer—A clinical perspective. JEADV Clin Pract 2024;4:145–55.

Abhishek K, Kawahara J, Hamarneh G. Predicting the clinical management of skin lesions using deep learning. Sci Rep 2021;11:7769.

Absci. ABS-201. https://www.absci.com/abs-201-case-study/; 2025. (accessed June 16, 2025).

Ahmad B, Usama M, Ahmad T, Khatoon S, Alam CM. An ensemble model of convolution and recurrent neural network for skin disease classification. Int J Imaging Syst Tech 2022;32:218–29.

Aijaz SF, Khan SJ, Azim F, Shakeel CS, Hassan U. Deep learning application for effective classification of different types of psoriasis. J Healthc Eng 2022;2022:7541583.

Ali W, Williams J, Xiong B, Zou J, Daneshjou R. Machine learning for early detection of hidradenitis suppurativa: a feasibility study using medical insurance claims data. JID Innov 2025;5:100362.

Barraza GA, Castro-Guijarro AC, De La Fuente Hoffmann V, Bolívar Ávila SJ, Flamini MI, Sanchez AM. Drug repositioning for rosacea disease: biological TARGET identification, molecular docking, pharmacophore mapping, and molecular dynamics analysis. Comput Biol Med 2024;181:108988.

Berna R, Mitra N, Hoffstad O, Wan J, Margolis DJ. Identifying phenotypes of atopic dermatitis in a longitudinal United States cohort using unbiased statistical clustering. J Invest Dermatol 2020;140:477–9.

Bettuzzi T, Hua C, Diaz E, Colin A, Wolkenstein P, De Prost N, et al. Epidermal necrolysis: characterization of different phenotypes using an unsupervised clustering analysis. Br J Dermatol 2022;186:1037–9.

Binol H, Plotner A, Sopkovich J, Kaffenberger B, Niazi MKK, Gurcan MN. Ros-NET: a deep convolutional neural network for automatic identification of rosacea lesions. Skin Res Technol 2020;26:413–21.

Capurro N, Pastore VP, Touijer L, Odone F, Cozzani E, Gasparini G, et al. A deep learning approach to direct immunofluorescence pattern recognition in autoimmune bullous diseases. Br J Dermatol 2024;191:261–6.

Cazzaniga S, Pezzolo E, Garcovich S, Naldi L. IRHIS Study Group. Late-onset hidradenitis suppurativa: a cluster analysis of the National Italian Registry IRHIS. J Am Acad Dermatol 2021;85:e29–32.

Chakraborty C, Achar U, Nayek S, Achar A, Mukherjee R. CAD-PsorNet: deep transfer learning for computer-assisted diagnosis of skin psoriasis. Sci Rep 2024;14:26557.

Chen JC, Dai Z, Christiano AM. Regulatory network analysis defines unique drug mechanisms of action and facilitates patient-drug matching in alopecia areata clinical trials. Comput Struct Biotechnol J 2021;19:4751–8.

Choi S, Kim J, Lee JH, Lee YN, Lee JH. Clinical and histological characteristics of localized morphea, generalized morphea and systemic sclerosis: a comparative study aided by machine learning. Acta Derm Venereol 2023;103:adv11953.

Clayton K, Vallejo A, Sirvent S, Davies J, Porter G, Reading IC, et al. Machine learning applied to atopic dermatitis transcriptome reveals distinct therapy-dependent modification of the keratinocyte immunophenotype. Br J Dermatol 2021;184:913–22.

Corbin A, Marques O. Assessing bias in skin lesion classifiers with contemporary deep learning and post-hoc explainability techniques. IEEE Access 2023;11:78339–52.

Czajkowska J, Badura P, Korzekwa S, Płatkowska-Szczerek A. Deep learning approach to skin layers segmentation in inflammatory dermatoses. Ultrasonics 2021a;114:106412.

Czajkowska J, Badura P, Korzekwa S, Płatkowska-Szczerek A. Automated segmentation of epidermis in high-frequency ultrasound of pathological skin using a cascade of DeepLab v3+ networks and fuzzy connectedness. Comput Med Imaging Graph 2022;95:102023.

Czajkowska J, Badura P, Korzekwa S, Płatkowska-Szczerek A, Słowińska M. Deep learning-based high-frequency ultrasound skin image classification with multicriteria model evaluation. Sensors (Basel) 2021b;21:5846.

Damiani G, Conic RRZ, Pigatto PDM, Carrera CG, Franchi C, Cattaneo A, et al. Predicting secukinumab fast-responder profile in psoriatic patients: advanced application of artificial-neural-networks (ANNs). J Drugs Dermatol 2020;19:1241–6.

Deng Z, Chen M, Zhao Z, Xiao W, Liu T, Peng Q, et al. Whole genome sequencing identifies genetic variants associated with neurogenic inflammation in rosacea. Nat Commun 2023;14:3958.

Dev K, Ho CJH, Bi R, Yew YW, S DU, Attia ABE, et al. Machine learning assisted handheld confocal Raman micro-spectroscopy for identification of clinically relevant atopic eczema biomarkers. Sensors (Basel) 2022;22:4674.

Dunlap GS, Billi AC, Xing X, Ma F, Maz MP, Tsoi LC, et al. Single-cell transcriptomics reveals distinct effector profiles of infiltrating T cells in lupus skin and kidney. JCI Insight 2022;7:e156341.

Eskandari A, Sharbatdar M. Efficient diagnosis of psoriasis and lichen planus cutaneous diseases using deep learning approach. Sci Rep 2024;14:9715.

Esteva A, Kuprel B, Novoa RA, Ko J, Swetter SM, Blau HM, et al. Dermatologist-level classification of skin cancer with deep neural networks. Nature 2017;542:115–8.

Fortino V, Wisgrill L, Werner P, Suomela S, Linder N, Jalonen E, et al. Machine-learning–driven biomarker discovery for the discrimination between allergic and irritant contact dermatitis. Proc Natl Acad Sci USA 2020;117:33474–85.

Franks JM, Martyanov V, Cai G, Wang Y, Li Z, Wood TA, et al. A machine learning classifier for assigning individual patients with systemic sclerosis to intrinsic molecular subsets. Arthritis Rheumatol 2019;71:1701–10.

Gao Y, Xu C, Yan Q. Genetic analyses reveal association between rosacea and cardiovascular diseases. Sci Rep 2025;15:6657.

Ge L, Li Y, Wu Y, Fan Z, Song Z. Differential diagnosis of rosacea using machine learning and dermoscopy. Clin Cosmet Investig Dermatol 2022;15:1465–73.

Ghosh D, Ding L, Sivaprasad U, Geh E, Biagini Myers J, Bernstein JA, et al. Multiple transcriptome data analysis reveals biologically relevant atopic dermatitis signature genes and pathways. PLoS One 2015;10:e0144316.

Giavina Bianchi M, D'adario A, Giavina Bianchi P. Machado BS; IChat Group. Three versions of an atopic dermatitis case report written by humans, artificial intelligence, or both: identification of authorship and preferences. J Allergy Clin Immunol Glob 2024;4:100373.

González-Manso A, Agut-Busquet E, Romaní J, Vilarrasa E, Bittencourt F, Mensa A, et al. Hidradenitis suppurativa: proposal of classification in two endotypes with two-step cluster analysis. Dermatology 2021;237:365–71.

Greenfield DA, Feizpour A, Evans CL. Quantifying inflammatory response and drug-aided resolution in an atopic dermatitis model with deep learning. J Invest Dermatol 2023;143:1430–8.e4.

Groh M, Badri O, Daneshjou R, Koochek A, Harris C, Soenksen LR, et al. Deep learning-aided decision support for diagnosis of skin disease across skin tones. Nat Med 2024;30:573–83.

Guo L, Yang Y, Ding H, Zheng H, Yang H, Xie J, et al. A deep learning-based hybrid artificial intelligence model for the detection and severity assessment of vitiligo lesions. Ann Transl Med 2022;10:590.

Gustafson E, Pacheco J, Wehbe F, Silverberg J, Thompson W. A machine learning algorithm for identifying atopic dermatitis in adults from electronic health records. Proc (IEEE Int Conf Healthc Inform) 2017 2017:83–90.

Hammad M, Pławiak P, ElAffendi M, El-Latif AAA, Latif AAA. Enhanced deep learning approach for accurate eczema and psoriasis skin detection. Sensors (Basel) 2023;23:7295.

Hart ML, Kato R, Rolauffs B. Editorial: artificial intelligence in predicting, determining and controlling cell phenotype or tissue function in inflammatory diseases. Front Immunol 2024;15:1443534.

Hillmer D, Merhi R, Boniface K, Taieb A, Barnetche T, Seneschal J, et al. Evaluation of facial vitiligo severity with a mixed clinical and artificial intelligence approach. J Invest Dermatol 2024;144:351–7.e4.

Ho CJH, Yew YW, Dinish US, Kuan AHY, Wong MKW, Bi R, et al. Handheld confocal Raman spectroscopy (CRS) for objective assessment of skin barrier function and stratification of severity in atopic dermatitis (AD) patients. J Dermatol Sci 2020;98:20–5.

Hu Z, Zheng M, Guo Z, Zhou W, Zhou W, Yao N, et al. Single-cell sequencing reveals distinct immune cell features in cutaneous lesions of pemphigus vulgaris and bullous pemphigoid. Clin Immunol 2024;263:110219.

Huang K, Jiang Z, Li Y, Wu Z, Wu X, Zhu W, et al. The classification of six common skin diseases based on Xiangya-derm: development of a Chinese database for artificial intelligence. J Med Internet Res 2021a;23:e26025.

Huang Y, Wen HJ, Guo YL, Wei TY, Wang WC, Tsai S-F, et al. Prenatal exposure to air pollutants and childhood atopic dermatitis and allergic rhinitis adopting machine learning approaches: 14-year follow-up birth cohort study. Sci Total Environ 2021b;777:145982.

Hurault G, Domínguez-Hüttinger E, Langan SM, Williams HC, Tanaka RJ. Personalized prediction of daily eczema severity scores using a mechanistic machine learning model. Clin Exp Allergy 2020;50:1258–66.

Jiang Z, Li J, Kong N, Kim JH, Kim BS, Lee MJ, et al. Accurate diagnosis of atopic dermatitis by combining transcriptome and microbiota data with supervised machine learning. Sci Rep 2022;12:290.

Jütte L, González-Villà S, Quintana J, Steven M, Garcia R, Roth B. Integrating generative AI with ABCDE rule analysis for enhanced skin cancer diagnosis, dermatologist training and patient education. Front Med (Lausanne) 2024;11:1445318.

Karthik R, Vaichole TS, Kulkarni SK, Yadav O, Khan F. Eff2Net: an efficient channel attention-based convolutional neural network for skin disease classification. Biomed Signal Process Control 2022;73:103406.

Khan Y, Todorov A, Torah R, Beeby S, Ardern-Jones MR. Skin sensing and wearable technology as tools to measure atopic dermatitis severity. Skin Health Dis 2024;4:e449.

Kim C, Gadgil SU, DeGrave AJ, Omiye JA, Cai ZR, Daneshjou R, et al. Transparent medical image AI via an image-text foundation model grounded in medical literature. Nat Med 2024;30:1154–65.

Kirby J, Kim K, Zivkovic M, Wang S, Garg V, Danavar A, et al. Uncovering the burden of hidradenitis suppurativa misdiagnosis and underdiagnosis: a machine learning approach. Front Med Technol 2024;6:1200400.

Koopman JJE, Buhler KA, Choi MY. From machine learning to clinical practice: phenotypic clusters of anti-MDA5 antibody-positive dermatomyositis. Brief Bioinform 2024;25:bbae303.

Lakdawala N, Channa L, Gronbeck C, Lakdawala N, Weston G, Sloan B, et al. Assessing the accuracy and comprehensiveness of ChatGPT in offering clinical guidance for atopic dermatitis and acne vulgaris. JMIR Dermatol 2023;6:e50409.

Li Q, Yang Z, Chen K, Zhao M, Long H, Deng Y, et al., Human-multimodal deep learning collaboration in 'precise' diagnosis of lupus erythematosus subtypes and similar skin diseases. J Eur Acad Dermatol Venereol 2024;38:2268–79.

Lin B, Xu Y, Bao X, Zhao Z, Wang Z, Yin J. SkinGEN: an explainable dermatology diagnosis-to-generation framework with interactive vision-language models. Cagliari, Italy: Paper presented at: IUI '25: 30th International Conference on Intelligent User Interfaces; 24–27 March 2025.

Liu L, Liang C, Xue Y, Chen T, Chen Y, Lan Y, et al. An intelligent diagnostic model for Melasma based on deep learning and multimode image input. Dermatol Ther (Heidelb) 2022b;13:569–79.

Liu S, Fan Y, Duan M, Wang Y, Su G, Ren Y, et al. AcneGrader: an ensemble pruning of the deep learning base models to grade acne. Skin Res Technol 2022a;28:677–88.

Liu Y, Jain A, Eng C, Way DH, Lee K, Bui P, et al. A deep learning system for differential diagnosis of skin diseases. Nat Med 2020;26:900–8.

Liu Y, Wang H, Taylor M, Cook C, Martínez-Berdeja A, North JP, et al. Classification of human chronic inflammatory skin disease based on single-cell immune profiling. Sci Immunol 2022c;7:eabl9165.

Klçoğlu MS, Yurdakul OV, Aydn T. Evaluation of chat generative pretrained transformer's responses to frequently asked questions about psoriatic arthritis: a study on quality and readability. Ann Med Res 2025;32:78–83.

Maintz L, Welchowski T, Herrmann N, Brauer J, Kläschen AS, Fimmers R, et al. Machine learning-based deep phenotyping of atopic dermatitis: severity-associated factors in adolescent and adult patients. JAMA Dermatol 2021;157:1414–24.

Mao R, Li J. Construction of a molecular diagnostic system for neurogenic rosacea by combining transcriptome sequencing and machine learning. BMC Med Genomics 2024;17:232.

Martínez BA, Shrotri S, Kingsmore KM, Bachali P, Grammer AC, Lipsky PE. Machine learning reveals distinct gene signature profiles in lesional and nonlesional regions of inflammatory skin diseases. Sci Adv 2022;8:eabn4776.

McDuff D, Schaekermann M, Tu T, Palepu A, Wang A, Garrison J, et al. Towards accurate differential diagnosis with large language models. Nature 2025;642:451–7.

McLeish E, Slater N, Mastaglia FL, Needham M, Coudert JD. From data to diagnosis: how machine learning is revolutionizing biomarker discovery in idiopathic inflammatory myopathies [published correction appears in Brief Bioinform 2024;25:bbae100] Brief Bioinform 2023;25:bbad514.

Mohan J, Sivasubramanian A, V S, Ravi V. Enhancing skin disease classification leveraging transformer-based deep learning architectures and explainable AI. Comput Biol Med 2025;190:110007.

Moon CI, Kim EB, Baek YS, Lee O. Transformer based on the prediction of psoriasis severity treatment response. Biomed Signal Process Control 2024;89:105743.

Muhaba KA, Dese K, Aga TM, Zewdu FT, Simegn GL. Automatic skin disease diagnosis using deep learning from clinical image and patient information. Skin Health Dis 2022;2:e81.

Muñoz-López C, Ramírez-Cornejo C, Marchetti MA, Han SS, Del Barrio-Díaz P, Jaque A, et al. Performance of a deep neural network in teledermatology: a single-centre prospective diagnostic study. J Eur Acad Dermatol Venereol 2021;35:546–53.

Nair S, Lushington GH, Purushothaman M, Rubin B, Jupe E, Gattam S. Prediction of lupus classification criteria via generative AI medical record profiling. BioTech (Basel) 2025;14:15.

Neely J, Hartoularos G, Bunis D, Sun Y, Lee D, Kim S, et al. Multi-modal single-cell sequencing identifies cellular immunophenotypes associated with juvenile dermatomyositis disease activity. Front Immunol 2022;13:902232.

Nicholas A, Spraul A, Fleischer AB Jr. Prescriber phenotypes: variability in topical rosacea treatment patterns among United States dermatologists. J Clin Med 2024;13:6275.

Okamoto T, Kawai M, Ogawa Y, Shimada S, Kawamura T. Artificial intelligence for the automated single-shot assessment of psoriasis severity. J Eur Acad Dermatol Venereol 2022;36:2512–5.

Park S, Chien AL, Lin B, Li K. FACES: a deep-learning-based parametric model to improve rosacea diagnoses. Appl Sci (Basel) 2023;13:970.

Patrick MT, Raja K, Miller K, Sotzen J, Gudjonsson JE, Elder JT, et al. Drug repurposing prediction for immune-mediated cutaneous diseases using a word-embedding-based machine learning approach. J Invest Dermatol 2018a;139:683–91.

Patrick MT, Stuart PE, Raja K, Gudjonsson JE, Tejasvi T, Yang J, et al. Genetic signature to provide robust risk assessment of psoriatic arthritis development in psoriasis patients. Nat Commun 2018b;9:4178.

Perez RK, Gordon MG, Subramaniam M, Kim MC, Hartoularos GC, Targ S, et al. Single-cell RNA-seq reveals cell type-specific molecular and genetic associations to lupus. Science 2022;376:eabf1970.

Pournara E, Kormaksson M, Nash P, Ritchlin CT, Kirkham BW, Ligozio G, et al. Clinically relevant patient clusters identified by machine learning from the clinical development programme of secukinumab in psoriatic arthritis. RMD Open 2021;7:e001845.

Prakash AV, Park JW, Seong JW, Kang TJ. Repositioned drugs for inflammatory diseases such as sepsis, asthma, and atopic dermatitis. Biomol Ther (Seoul) 2020;28:222–9.

Prasannanjaneyulu V, Nene S, Jain H, Nooreen R, Otavi S, Chitlangya P, et al. Old drugs, new tricks: emerging role of drug repurposing in the management of atopic dermatitis. Cytokine Growth Factor Rev 2022;65:12–26.

Queiro R, Seoane-Mato D, Laiz A, Agirregoikoa EG, Montilla C, Park HS, et al. Minimal disease activity (MDA) in patients with recent-onset psoriatic arthritis: predictive model based on machine learning. Arthritis Res Ther 2022;24:153.

Rajalingam K, Levin N, Marques O, Grichnik J, Lin A, Chen WS. Treatment options and emotional well-being in patients with rosacea: an unsupervised machine learning analysis of over 200,000 posts. JAAD Int 2023;13:172–8.

Schaap MJ, Cardozo NJ, Patel A, De Jong EMGJ, Van Ginneken B, Seyger MMB. Image-based automated Psoriasis Area Severity Index scoring

by Convolutional Neural Networks. J Eur Acad Dermatol Venereol 2022;36:68–75.

Seremet T, Di Domizio J, Girardin A, Yatim A, Jenelten R, Messina F, et al. Immune modules to guide diagnosis and personalized treatment of inflammatory skin diseases. Nat Commun 2024;15:10688.

Shen Y, Li H, Sun C, Ji H, Zhang D, Hu K, et al. Optimizing skin disease diagnosis: harnessing online community data with contrastive learning and clustering techniques. NPJ Digit Med 2024;7:28.

Shrivastava VK, Londhe ND, Sonawane RS, Suri JS. A novel approach to multiclass psoriasis disease risk stratification: machine learning paradigm. Biomed Signal Process Control 2016;28:27–40.

Shrivastava VK, Londhe ND, Sonawane RS, Suri JS. A novel and robust Bayesian approach for segmentation of psoriasis lesions and its risk stratification. Comput Methods Programs Biomed 2017;150:9–22.

Sulejmani P, Negris O, Aoki V, Chu CY, Eichenfield L, Misery L, et al. A large language model artificial intelligence for patient queries in atopic dermatitis. J Eur Acad Dermatol Venereol 2024;38:e531–5.

T A, Abhishek S, S R. A novel framework for rosacea detection using Swin Transformers and explainable artificial intelligence. Alex Eng J 2025;118:36–58.

Tao Y, Hua G, Min S, Xiao-Hong L, Jia-Hui J, Biao T, et al. Verification of biological markers of subacute cutaneous lupus erythematosus via TMT labelling proteomics combined with transcriptome data. Ann Med 2025;57:2500696.

Tomalin LE, Kim J, Correa Da Rosa J, Lee J, Fitz LJ, Berstein G, et al. Early quantification of systemic inflammatory proteins predicts long-term treatment response to tofacitinib and etanercept. J Invest Dermatol 2020;140:1026–34.

Vodrahalli K, Daneshjou R, Novoa RA, Chiou A, Ko JM, Zou J. TrueImage: a machine learning algorithm to improve the quality of telehealth photos. Pac Symp Biocomput 2021;26:220–31.

Wang A, Fulton R, Hwang S, Margolis DJ, Mowery D. Patient phenotyping for atopic dermatitis with transformers and machine learning: algorithm development and validation study. JMIR Form Res 2024b;8:e52200.

Wang D, Pang N, Wang Y, Zhao H. Unlabeled skin lesion classification by self-supervised topology clustering network. Biomed Signal Process Control 2021b;66:102428.

Wang J, Luo L, Ding Q, Wu Z, Peng Y, Li J, et al. Development of a multi-target strategy for the treatment of vitiligo via machine learning and network analysis methods. Front Pharmacol 2021a;12:754175.

Wang X, Hu H, Yan G, Zheng B, Luo J, Fan J. Identification and validation of interferon-stimulated gene 15 as a biomarker for dermatomyositis by integrated bioinformatics analysis and machine learning. Front Immunol 2024c;15:1429817.

Wang Y, Chen J, Shen ZY, Zhang J, Zhu YJ, Xia XQ. Screening of diagnostic biomarkers and immune infiltration characteristics linking rheumatoid arthritis and rosacea based on bioinformatics analysis. J Inflamm Res 2024a;17:5177–95.

Wang Y, Qin D, Jin L, Liang G. Caffeoyl malic acid is a potential dual inhibitor targeting TNFα/IL-4 evaluated by a combination strategy of network analysis-deep learning-molecular simulation. Comput Biol Med 2022;145:105410.

Wiala A, Ranjan R, Schnidar H, Rappersberger K, Posch C. Automated classification of hidradenitis suppurativa disease severity by convolutional neural network analyses using calibrated clinical images. J Eur Acad Dermatol Venereol 2024;38:576–82.

Wu H, Yin H, Chen H, Sun M, Liu X, Yu Y, et al. A deep learning-based smartphone platform for cutaneous lupus erythematosus classification assistance: simplifying the diagnosis of complicated diseases. J Am Acad Dermatol 2021;85:792–3.

Wu H, Yin H, Chen H, Sun M, Liu X, Yu Y, et al. A deep learning, image based approach for automated diagnosis for inflammatory skin diseases. Ann Transl Med 2020b;8:581.

Wu JJ, Hong CH, Merola JF, Gruben D, Güler E, Feeney C, et al. Predictors of nonresponse to dupilumab in patients with atopic dermatitis: a machine learning analysis. Ann Allergy Asthma Immunol 2022a;129:354–9.e5.

Wu W, Chen G, Zhang Z, He M, Li H, Yan F. Construction and verification of atopic dermatitis diagnostic model based on pyroptosis related biological markers using machine learning methods. BMC Med Genomics 2023;16:138.

Wu X, Xiao Y, Ma J, Wang A, Circular RNA. a novel potential biomarker for skin diseases. Pharmacol Res 2020a;158:104841.

Xiong J, Chen G, Liu Z, Wu X, Xu S, Xiong J, et al. Construction of regulatory network for alopecia areata progression and identification of immune monitoring genes based on multiple machine-learning algorithms. Precis Clin Med 2023;6:pbad009.

Xu X, Zhang HY. The immunogenetics of psoriasis and implications for drug repositioning. Int J Mol Sci 2017;18:2650.

Xue Y, Zhang J, Li C, Liu X, Kuang W, Deng J, et al. Machine learning for screening and predicting the risk of anti-MDA5 antibody in juvenile dermatomyositis children. Front Immunol 2023;13:940802.

Yan S, Yu Z, Primiero C, Vico-Alonso C, Wang Z, Yang L, et al. A multimodal vision foundation model for clinical dermatology. Nat Med 2025b;31:2691–702.

Yang AF, Patel S, Chun KS, Richards D, Walter JR, Okamoto K, et al. Artificial intelligence-enabled wearable devices and nocturnal scratching in mild atopic dermatitis. JAMA Dermatol 2025;161:406–10.

Yang Y, Guo L, Wu Q, Zhang M, Zeng R, Ding H, et al. Construction and evaluation of a deep learning model for assessing acne vulgaris using clinical images. Dermatol Ther (Heidelb) 2021;11:1239–48.

Yu K, Syed MN, Bernardis E, Gelfand JM. Machine learning applications in the evaluation and management of psoriasis: a systematic review. J Psoriasis Psoriatic Arthritis 2020;5:147–59.

Yu Z, Kaizhi S, Jianwen H, Guanyu Y, Yonggang W. A deep learning-based approach toward differentiating scalp psoriasis and seborrheic dermatitis from dermoscopic images. Front Med (Lausanne) 2022;9:965423.

Zhang S, Chang M, Zheng L, Wang C, Zhao R, Song S, et al. Deep analysis of skin molecular heterogeneities and their significance on the precise treatment of patients with psoriasis. Front Immunol 2024a;15:1326502.

Zhang T, Nie Y. Prediction of the risk of alopecia areata progressing to alopecia totalis and alopecia universalis: biomarker development with bioinformatics analysis and machine learning. Dermatology 2022;238:386–96.

Zhang Y, Chen R, Nguyen D, Choi S, Gabel C, Leonard N, et al. Assessing the ability of an artificial intelligence chatbot to translate dermatopathology reports into patient-friendly language: a cross-sectional study. J Am Acad Dermatol 2024b;90:397–9.

Zhang Y, Hu Y, Li K, Pan X, Mo X, Zhang H. Exploring the influence of transformer-based multimodal modeling on clinicians' diagnosis of skin diseases: a quantitative analysis. Digit Health 2024c;10:20552076241257087.

Zhao S, Xie B, Li Y, Zhao X, Kuang Y, Su J, et al. Smart identification of psoriasis by images using convolutional neural networks: a case study in China. J Eur Acad Dermatol Venereol 2020;34:518–24.

Zhao Z, Wu CM, Zhang S, He F, Liu F, Wang B, et al. A novel convolutional neural network for the diagnosis and classification of rosacea: usability study. JMIR Med Inform 2021;9:e23415.

Zhong F, He K, Ji M, Chen J, Gao T, Li S, et al. Optimizing vitiligo diagnosis with ResNet and Swin transformer deep learning models: a study on performance and interpretability. Sci Rep 2024;14:9127.

Zhou J, He X, Sun L, Xu J, Chen X, Chu Y, et al. Pre-trained multimodal large language model enhances dermatological diagnosis using SkinGPT-4. Nat Commun 2024a;15:5649.

Zhou X, Zhou H, Luo X, Wu RF. Discovery of biomarkers in the psoriasis through machine learning and dynamic immune infiltration in three types of skin lesions. Front Immunol 2024b;15:1388690.

Zhu CY, Wang YK, Chen HP, Gao KL, Shu C, Wang JC, et al. A deep learning based framework for diagnosing multiple skin diseases in a clinical environment. Front Med (Lausanne) 2021b;8:626369.

Zhu JL, Tran LT, Smith M, Zheng F, Cai L, James JA, et al. Modular gene analysis reveals distinct molecular signatures for subsets of patients with cutaneous lupus erythematosus. Br J Dermatol 2021a;185:563–72.